

PHYSIOLOGY

(Physiology)

This science studies how each organ of the body performs its functions so that the whole body lives a normal life free from diseases.

The human body is composed of a group of systems, such as the circulatory system, digestive system, nervous system, reproductive system, etc. Each system is composed of a group of organs. Each organ complements the function of the other. For example, the digestive system consists of the mouth, then the stomach, then the intestines (small and large), and also includes other accessory organs like salivary glands, liver, and pancreas. Each organ is composed of a group of tissues, and the unit that makes up tissue is the cell.

The Cell:

It is the unit of life for the bodies of all living organisms, whether plant or animal. The cell consists of a living substance with a gelatinous consistency called "protoplasm." The protoplasm condenses outside the cell forming a thin membrane known as the cell wall. This wall determines the shape of the cell and also regulates the exchange of nutrients and gases between the inside and outside of the cell.

Functional Structure of the Cell:

1. Cell Membranes:

Called so because they are not limited to the outer wall but extend to surround the different organelles inside the cell.

2. Endoplasmic Reticulum (ER):

A network connected to the nucleus and the outside of the cell. It is of two types:

a. **Rough ER (Granular):** Contains ribonucleic acid (RNA) and is responsible for protein synthesis in the cell.

b. **Smooth ER (Agranular):** Does not contain this acid and is responsible for lipid formation.

3. Golgi Apparatus (Body):

Also an endoplasmic network connected to the endoplasmic reticulum, notably found in secretory cells.

Functions:

1. Temporary storage of substances formed in the endoplasmic reticulum.

2. Formation of carbohydrates.

3. Formation of lysosomes.

4. Mitochondria:

A fundamental organelle in the cell where the combustion of nutrients occurs, releasing energy necessary for the cell.

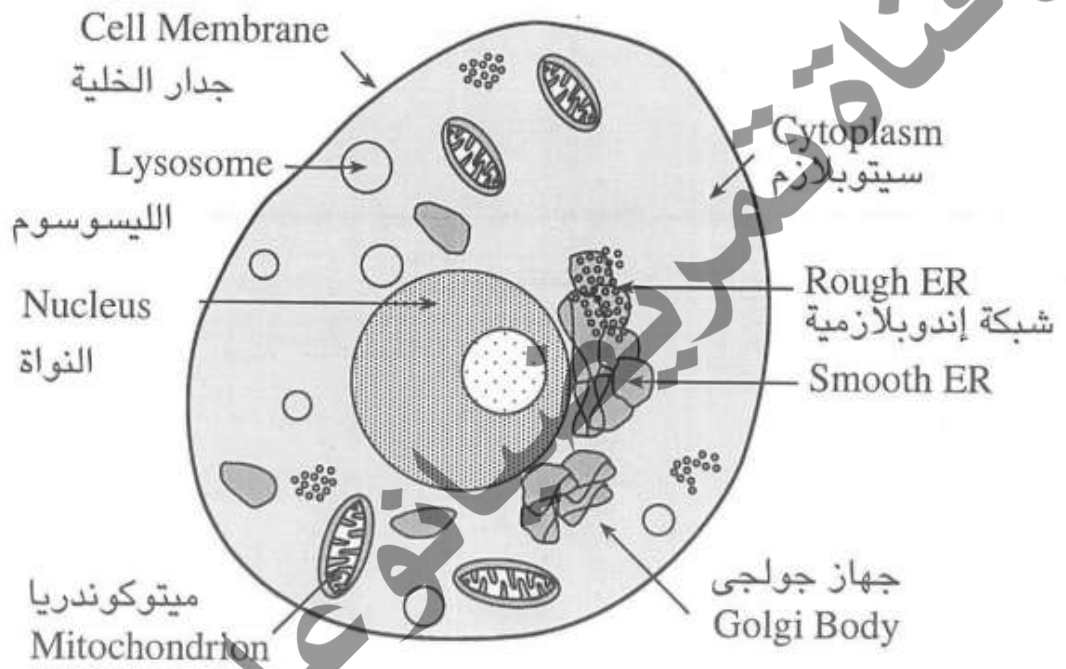
5. Lysosomes:

They form the digestive system of the cell. They are small bodies surrounded by an outer membrane and contain digestive enzymes that attack germs and waste inside the cell.

6. Nucleus:

It is the control center of the cell, both for chemical reactions and for cell reproduction. This is due to the nucleus containing deoxyribonucleic acid (DNA), which contains the genes that determine the types of different enzymes in the cell, thus influencing chemical processes. On the other hand, these genes themselves replicate, leading to cell division to form two cells, and so on.

The Cell



REVIEW QUESTIONS: THE CELL

1. Label the parts of the cell in the diagram?
2. Discuss the function of each:
 - a. Cell membranes
 - b. Endoplasmic reticulum
 - c. Nucleus
3. Complete the following:
 - a. Mitochondria are essential organelles in the cell, where inside them occurs
 - b. Lysosomes in the cell are responsible for
 - c. Functions of the Golgi apparatus include: 1. 2. 3.
4. Choose the correct answer from the following:
 - a. The combustion of nutrients and release of necessary energy occurs inside:
☐ The nucleus.
☐ The Golgi apparatus.
☐ The mitochondria.
☐ The lysosomes.

THE MUSCULAR SYSTEM

Importance of Muscles:

Muscles constitute about 40% of body weight, and their primary function is contraction. Through their contraction, they cause movement of the whole body or a part of it, like a limb. The functions and value of muscles become clearly apparent when they are paralyzed; the person becomes unable to eat, clean themselves, or dress. They also cannot speak, defend themselves, or avoid dangers. Undoubtedly, one of the most important functions of muscles is producing heat energy inside the body, helping to maintain a constant temperature in cold weather. We find that the breathing process occurs due to the contraction of respiratory muscles; if these are paralyzed, death occurs immediately. The balance of muscle contraction in the body allows for maintaining different body postures.

Types of Muscles:

1. Voluntary
- 2) Involuntary (Smooth)
- 3) Cardiac muscle

Changes Accompanying Muscle Contraction:

The contraction of voluntary muscles is accompanied by three types of changes:

1. Chemical changes.
2. Thermal changes.
3. Mechanical changes.

Chemical Changes of Muscle Contraction:

Detailed chemical studies of the muscle protein (actomyosin) have shown that its molecules exist in two forms: a relaxed form (when the muscle is relaxed) and a contracted form (when the muscle is contracted).

Thermal Changes in Muscle Contraction:

Muscle contraction is accompanied by the release of an amount of heat caused by the chemical reactions occurring in the muscle. It has been found that heat is released both during muscle contraction and during its relaxation, with the heat of contraction being approximately equal to the heat of relaxation.

Mechanical Changes of Muscle Contraction:

Muscle contraction is affected by several factors:

1. The degree of stretch of the muscle fibers.
2. The rate of muscle contraction.
3. Muscle fatigue.

Comparison between the three types of muscles

#	Voluntary Muscles	Involuntary Muscles (Smooth)	Cardiac Muscle
1	Attached to bones, voluntary	Found in viscera, involuntary	In the thoracic cavity
2	~40% of body weight	~10% of body weight	~500 grams
3	Cells are striated L & W under microscope	Cells are non-striated	Cells are striated
4	Supplied by voluntary nerves	Supplied by involuntary nerves (Sympathetic & Parasympathetic)	Supplied by involuntary nerves
5	Nerve severance leads to complete paralysis	Nerve severance does not paralyze the muscle	Nerve severance does not paralyze the heart muscle

QUESTIONS:

1. State the benefits of muscles.
 2. What are the changes that accompany muscle contraction?
 3. Compare the three types of muscles.
 4. Choose the correct answer from the following:
 - Characteristics of voluntary muscles include:
 - They constitute 40% of body weight.
 - They are supplied by sympathetic nerves.
 - They are not affected by nerve severance.
 - They are non-striated under the microscope.
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THE NERVOUS SYSTEM

It is the system that controls the different body systems to regulate processes necessary for life, so that each element performs its function at the right time and in a manner suitable for the benefit of the body. It includes:

(1) Central Nervous System (CNS):

- Brain
- Spinal cord

(2) Peripheral Nervous System (PNS):

- Cranial nerves – Spinal nerves – Nerves of the autonomic system.

The functional unit of the nervous system is the neuron, which is characterized by having extensions called nerve fibers that transmit sensations and nerve commands to and from the central nervous system.

Functions of the Spinal Cord:

- Transmits various signals and sensations from body parts like skin, muscles, and joints via sensory nerves to the higher central centers, and also transmits commands from the brain to various body organs like muscles.
- The spinal cord is an important center for reflex nervous actions.
- The spinal cord works on nourishing muscles and body tissues.

Reflex Action:

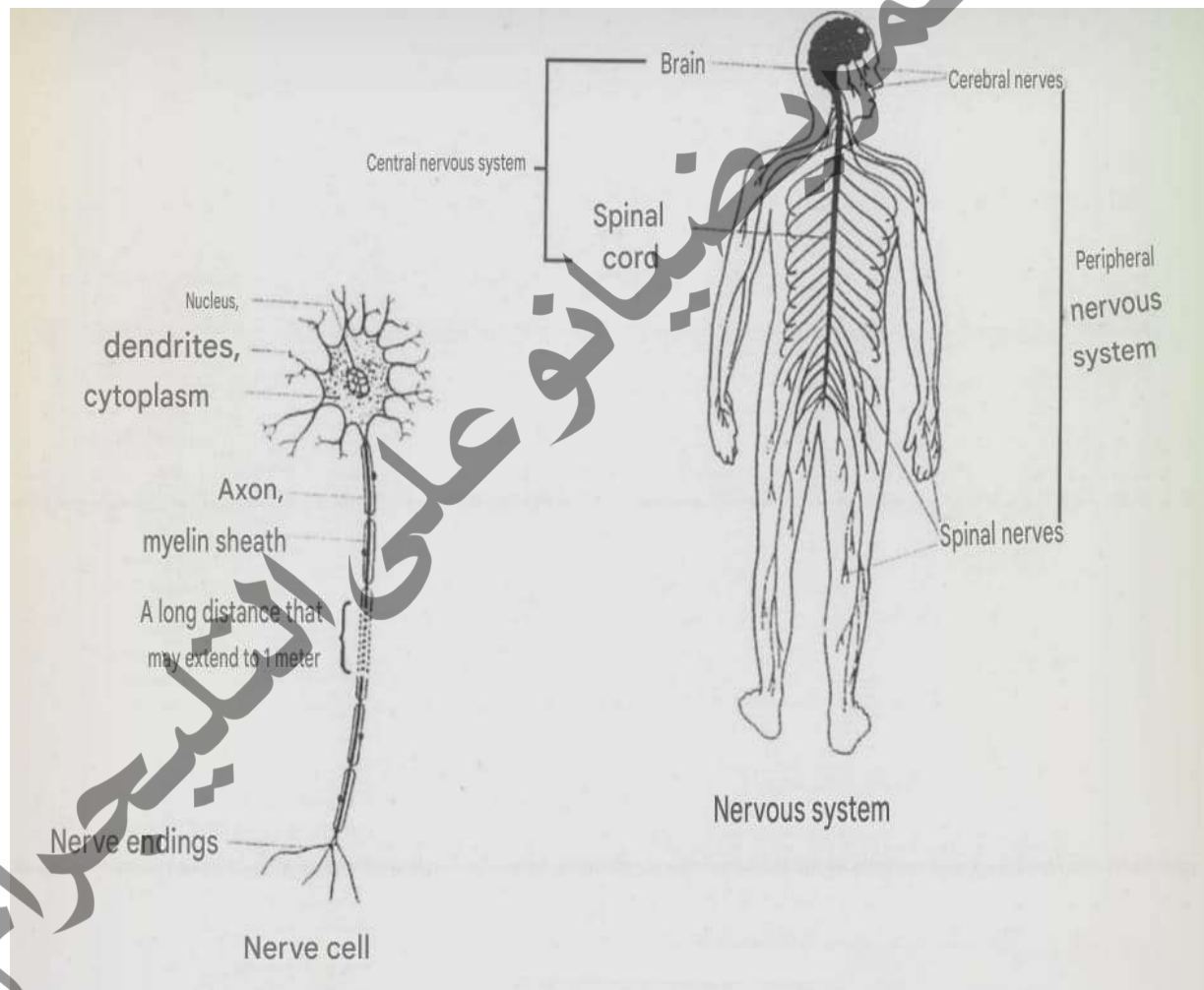
- It is an involuntary but necessary action for life, as it is an essential response:
 - To a natural need, such as regulating heartbeats, breathing movements, saliva secretion, and other ordinary involuntary processes.
 - To adapt a normal action to emergency circumstances, such as constricting the pupil when exposed to bright light or increasing heart rate during hemorrhage.
 - To a necessary urgent need to protect the body from harm or injury, such as contraction of the arm muscles if the hand touches a hot substance.

The Reflex Action

The reflex action works in voluntary and involuntary muscles and in gland secretion. A reflex action requires the presence of:

- A sensory receptor unit (receptor organ).
- A sensory nerve ascending to the central nervous system (brain or spinal cord).
- A nerve center in the brain or spinal cord to interpret incoming sensory signals and issue motor commands.
- A motor or secretory nerve descending from the central nervous system.
- An effector unit, such as muscles or glands.

The Nervous System



Cerebrospinal Fluid (CSF):

It is a clear fluid that surrounds the brain and spinal cord. It is found in the cavities within the brain substance and in the subarachnoid space surrounding the brain and spinal cord. The volume of cerebrospinal fluid is about 130 cm³ in an adult, with a specific gravity of 1.005. It contains protein at a rate of 20-40 mg% and is not coagulable. It contains glucose and urea in proportions similar to their presence in blood plasma, as well as a small number of lymphocytes in the normal state.

The elements of the CSF change in certain diseases, so examining this fluid has diagnostic value for some diseases.

The CSF acts as a soft cushion for the brain and spinal cord, protecting them from impacts and violent movements.

The Autonomic (Vegetative) Nervous System

- **Components:**

- Sympathetic Nervous System: Structure – Functions
- Parasympathetic Nervous System: Structure – Functions
- This system controls and regulates the body's vital functions, such as gland secretion, regulation of heartbeats, and movement and secretions of the digestive system, without the person being aware of or controlling them.

The autonomic nervous system is divided into two groups:

- The Sympathetic Nervous System.
- The Parasympathetic Nervous System.

First: The Sympathetic Nervous System:

Consists of two nerve cords extending from the base of the skull to the coccyx bones on either side of the vertebral column. Each cord has nodes called sympathetic ganglia. These ganglia exchange fibers with the spinal nerves.

Functions of the Sympathetic Nervous System:

1. Dilates the pupil and contracts the levator palpebrae superioris muscle (raising the upper eyelid).
2. Increases heart rate and raises blood pressure.
3. Dilates the bronchi and reduces their secretions.
4. Relaxes the muscles of the stomach and intestines and contracts the sphincter muscles.
5. Decreases secretions of various digestive glands and breaks down liver glycogen.
6. Relaxes the bladder muscles and contracts its sphincter muscles.
7. Stimulates the adrenal medulla to secrete adrenaline.
8. Increases blood flow to voluntary muscles and improves their performance.

Sympathetic nervous system activity increases during emotional excitement, physical exercise, hemorrhage, and exposure to extreme cold. All these conditions require an increase in the body's resistance.

Second: The Parasympathetic Nervous System:

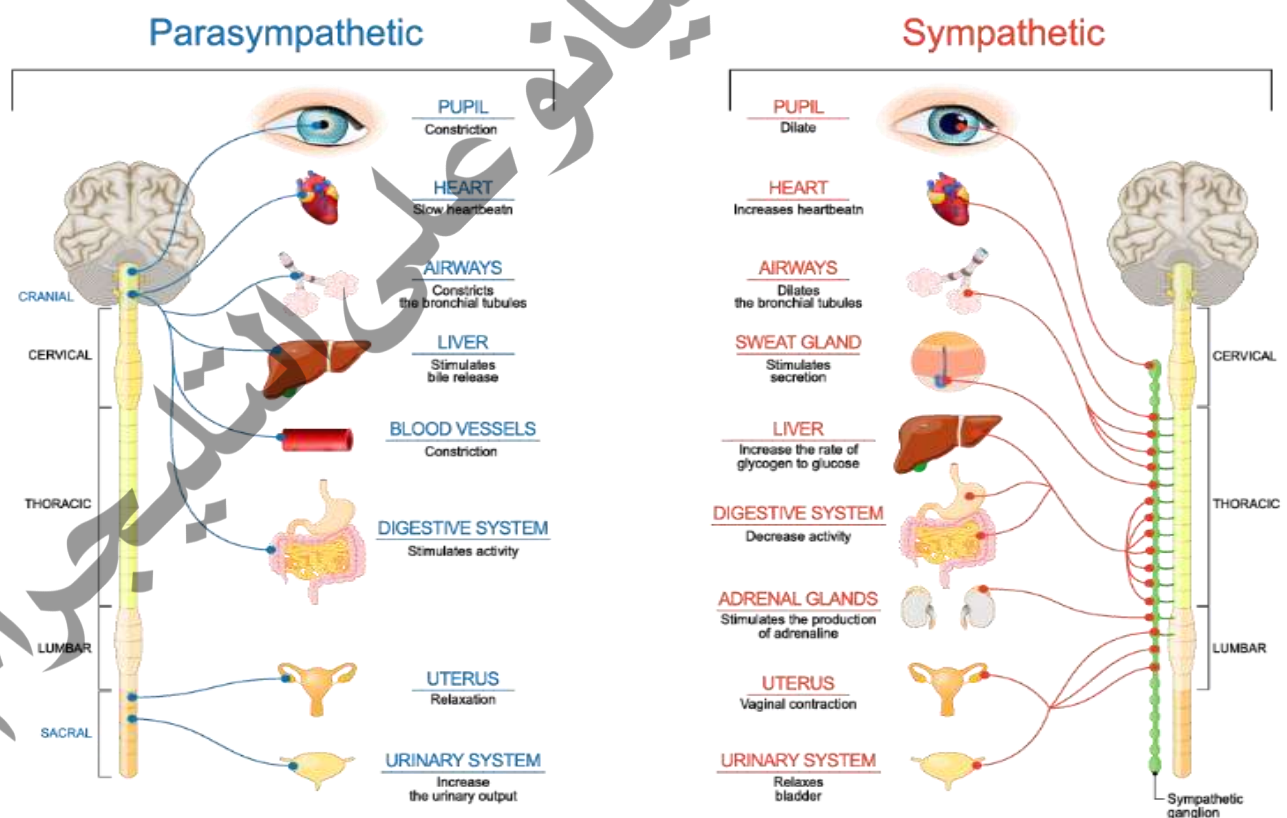
A group of nerves originating from the central nervous system in two regions:

- The group connected to cranial nerves 3, 7, 9, 10.
- The group connected to the spinal cord in the sacral region.

Functions of the Parasympathetic Nervous System:

1. Constricts the pupil and lowers the upper eyelid.
2. Decreases heart rate.
3. Constricts the bronchi and increases their secretions.
4. Contracts intestinal muscles and relaxes sphincter muscles.
5. Contracts the bladder muscles and relaxes the bladder sphincter muscles.
6. Assists in the secretion of various digestive juices.

Parasympathetic nervous system activity increases during rest and sleep. **Diagrams of Autonomic Nervous System**



REVIEW QUESTIONS: NERVOUS SYSTEM

1. Label the components of the reflex arc in the diagram.
2. List 5 functions of the sympathetic nervous system.
3. List 5 functions of the parasympathetic nervous system.
4. Complete the following:
 - a. is the functional unit of the nervous system, characterized by having that help with
 - b. Functions of the spinal cord include transmitting
 - c. Cerebrospinal fluid is essential for the brain because it
 - d. Sympathetic nervous system activity increases during, while parasympathetic nervous system activity increases during
5. Choose the correct answer from the following:
 - a. The sympathetic nervous system improves the performance of all these organs except:
 1. The heart.
 2. The adrenal gland.
 3. The digestive system.
 4. Voluntary muscles.
 - b. Characteristics of the parasympathetic nervous system include:
 1. It is active during rest and sleep.
 2. It causes dilation of the bronchi.
 3. It decreases intestinal motility.
 4. It increases the visual field of the eye.

THE CIRCULATORY SYSTEM

The circulatory system is responsible for the circulation of blood throughout the different parts of the body. It consists of the heart, arteries, blood capillaries, and veins. It is customary to divide blood circulation into two main parts:

Lesser (Pulmonary) Circulation:

Consists of the right side of the heart and the vessels in the lungs.

Greater (Systemic) Circulation:

Consists of the left side of the heart and all the blood vessels that pass through other parts of the body.

In both circuits, the heart acts as a pump moving blood through its vessels. Since the systemic circulation pumps blood to all parts of the body, the left side of the heart bears a greater burden and is therefore stronger and thicker than its right side.

The heart muscle, like any part of the body, cannot continue its activity unless it receives a sufficient amount of oxygen. This happens through blood carried by a special artery to the heart muscle, the "coronary artery." Narrowing or blockage of this artery deprives the heart muscle of oxygen, disrupting heart contractions and causing severe tightness and pain in the chest. This pain may extend to the shoulder or left arm. This condition is called "angina pectoris."

Nerves of the Heart:

The heart is supplied by nerves from the autonomic nervous system. These nerves are divided into two parts:

- **Sympathetic Nerves:**

- Supply all parts of the heart. Their functions:
 - Stimulate all properties of the heart muscle, increasing heart rate, contraction force, and speed of impulse conduction.
 - Dilate the coronary arteries, increasing the amount of blood passing through the heart.
 - Increase the heart's oxygen consumption.
- Sympathetic nerves are stimulated in the following conditions:
 - (a) Emotional states like fear, joy, anger, etc.
 - (b) Physical exertion, fasting, exposure to very cold weather, hemorrhage. All these conditions require an increase in the body's resistance.

- **Parasympathetic Nerves (Vagus Nerve):**

Supply the atria, the sinoatrial (SA) and atrioventricular (AV) nodes, and the bundle of His, but do not supply the ventricular muscle. Their functions:

- Reduce all properties of the heart muscle, slowing its rate and decreasing the force of its contraction (hence they do not supply the ventricles because a very slow rate would be detrimental).
- Constrict the caliber of the coronary arteries, reducing the amount of blood reaching the muscle.

Heart Rate:

Normal heart rate ranges between 60-100 beats per minute, averaging 72 beats per minute.

Several factors affect heart rate, such as age, body temperature, and athletic habits.

Heart rate is controlled by a center located in the medulla oblongata. The most important factors affecting the heart rate center are:

- (1) **Cardiac Nerves:** Stimulation of the sympathetic nerve, as in fear or during physical effort, causes an increase in heart rate. Stimulation of the vagus nerve, as during sleep, causes a slowing of the heart rate.
- (2) **Emotional Reactions:** Heart rate increases with joy, sadness, fear, embarrassment, etc., due to the effect of the emotional center in the hypothalamus.
- (3) **Blood Temperature:** An increase in blood temperature by one degree Celsius causes an increase of about 10 beats per minute.
- (4) **Amount of Blood Returning to the Heart:** As the amount of blood returning to the heart in the veins increases, the heart rate increases.
- (5) **Muscle Contraction:** When a muscle contracts, signals are emitted from the nerve endings within it that enter the spinal cord and ascend to the heart centers in the medulla oblongata, stimulating the center that causes an increase in heart rate.
- (6) **Blood Gases:** Heart rate increases if oxygen decreases or carbon dioxide increases in the blood, as occurs in cases of suffocation.
- (7) **Hormonal Effects:** Heart rate increases due to increased secretion of the thyroid hormone (located in the neck) or increased secretion of the hormone from the adrenal medulla.
- (8) **Physical Exertion:** Heart rate increases during physical exertion due to several factors: exertion causes a rise in blood temperature, an increase in the amount of blood returning to the heart, a decrease in oxygen and an increase in carbon dioxide in the blood, and also, signals stimulating the heart rate center are issued from the contracting muscles.

Arterial Blood Pressure:

It is the lateral pressure exerted by blood on the wall of the arteries. This pressure fluctuates with each contraction of the heart. It rises during ventricular contraction (systolic pressure) and falls during ventricular relaxation (diastolic pressure). Average systolic blood pressure is 120 mmHg and diastolic is 80 mmHg.

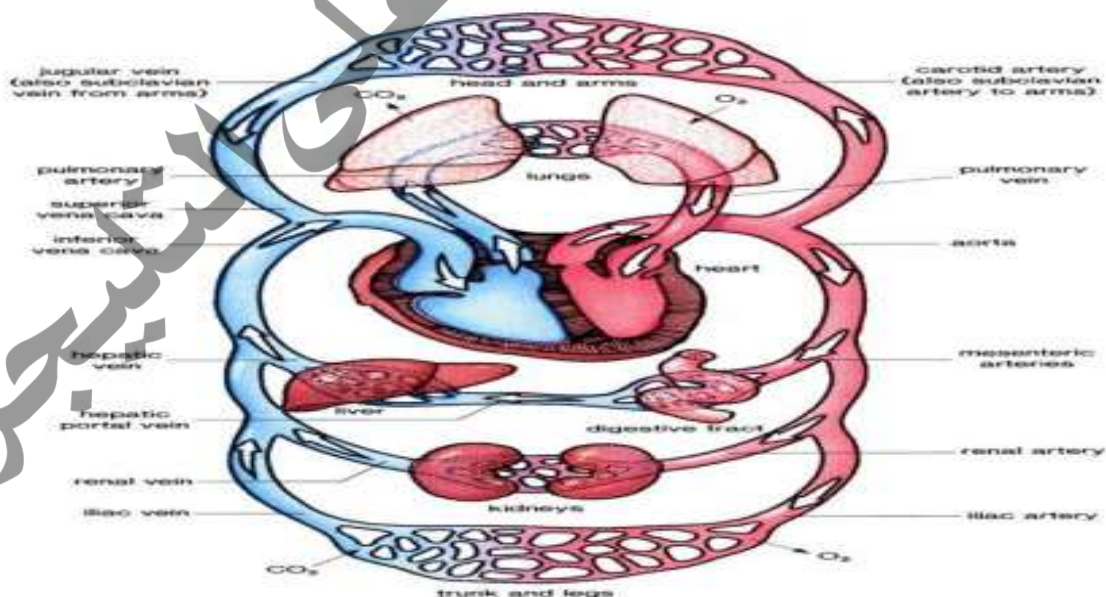
Normal Variations in Blood Pressure:

1. **Age:** Blood pressure changes with age. In a newborn, it is 50 systolic / 30 diastolic. After a few weeks, it becomes 60/90, reaching 120/80 by puberty.
2. **Sex:** Arterial blood pressure is lower in females than in males until menopause, after which blood pressure in females sometimes rises to a level higher than in males.
3. **Body Build:** In an obese person, blood pressure is higher than in a thin person.
4. **Eating:** Pressure rises slightly after eating because the amount of blood returning to the heart increases, thus increasing cardiac output.
5. **Physical Exertion:** Pressure rises temporarily during physical exertion.
6. **Emotional States:** Pressure rises by about 30 mmHg during emotional states.
7. **Sleep:** Pressure decreases during normal, calm sleep if not accompanied by disturbing dreams. When measuring blood pressure, it is measured from an artery in the arm at heart level because blood pressure is affected by gravity; in the arteries of the leg, it is higher than at heart level.

Factors Controlling Arterial Blood Pressure:

1. Cardiac Output: Blood pressure drops in cases of heart failure.
2. Peripheral Resistance encountered by blood during its passage through the arteries: Blood pressure rises with arterial constriction.
3. Elasticity of the arterial wall: Blood pressure rises in cases of arterial sclerosis.
4. Blood volume relative to circulatory volume: Blood pressure drops in cases of hemorrhage.

Diagram of Blood Circulation



The Lymphatic System

The lymphatic system consists of lymphatic vessels, lymph nodes, and lymph fluid.

Tissue Fluids and Lymph:

Cells live surrounded by a fluid that filters from blood plasma through the capillaries, carrying oxygen and nutrients to the tissues, and returns to the capillaries carrying carbon dioxide and waste products to the blood. This fluid constantly changes, but its chemical composition is stable.

The composition of tissue fluids is influenced by several factors:

- Capillary permeability.
- Blood pressure inside the capillary.
- Osmotic pressure of plasma proteins.

Lymphatic Circulation:

This circulation begins with capillaries similar to blood capillaries. The capillaries collect with each other to form lymphatic vessels resembling veins. These latter vessels collect until they end in the superior vena cava.

Function of the Lymphatic Circulation:

- Acts as a drain into which excess tissue fluid is poured.
- Lymph nodes... are collections of lymphatic cells located along the course of lymphatic vessels. These nodes act as filters, absorbing microbes and harmful substances from the lymph, preventing their passage into the blood.
- The cells of the lymph nodes produce white blood cells (lymphocytes).

Edema:

Edema is an abnormal increase in tissue fluid. Edema is dangerous, especially if it forms in the lungs (preventing gas exchange between blood and air) or in the brain (putting pressure on its centers). Also, fluid accumulation in any tissue creates pressure.

Types of Edema:

1. **Cardiac Edema:** In heart failure, blood pools in the veins, raising the pressure inside them, and consequently raising the pressure in the venous end of the capillary, leading to filtration of fluid from inside it into the tissues. Cardiac edema forms in right heart failure in the legs, and ascites (fluid in the abdomen) also occurs. In left heart failure, edema forms in the lungs.
2. **Renal Edema:** In chronic kidney infections, the body loses some plasma proteins in the urine, decreasing the osmotic pressure of plasma proteins, leading to edema.
3. **Nutritional Edema:** This type results from a deficiency of protein substances in the diet, leading to a deficiency in plasma proteins and a decrease in their osmotic pressure.
4. **Increased Capillary Permeability:** Such as in cases of poisoning with histamine-like substances.
5. **Localized Edema:** Results from external pressure on veins, as occurs in pregnancy, or internal vein blockage by a clot, or as in cases of ascites accompanying schistosomiasis and liver cirrhosis, where the portal vein is compressed by fibrosis, raising the pressure inside the vein, causing an increase in pressure in the venous end of the capillary, preventing the return of filtered fluid to the blood.
6. **Edema resulting from Blockage of Lymphatic Vessels:** As in elephantiasis, where worms enter and block the lymphatic vessels, preventing the return of lymph to the blood.

REVIEW QUESTIONS: CIRCULATION & EDEMA

1. What is meant by each of the following:
 - a. Arterial blood pressure.
 - b. Edema.
 - c. Angina pectoris.

2. Discuss:
 - a. The nerves of the heart and their functions.
 - b. Types of edema.

3. Complete the following:
 - a. The average heart rate per minute is, and the heart rate is regulated by a center located in, and the function of this center is affected by several factors including
 - b. Arterial blood pressure is controlled by some factors like
 - c. Arterial blood pressure changes due to

4. Choose the correct answer from the following:
 - a. The parasympathetic nerves that supply the heart:
 - () Increase its rate.
 - () Decrease the force of its contraction.
 - () Dilate the coronary arteries.
 - () Increase the heart's oxygen consumption.

 - b. Arterial blood pressure decreases during:
 - () Physical exertion.
 - () Emotional states.
 - () After eating.
 - () During sleep.

 - c. Edema that occurs in elephantiasis results from:
 - () Malnutrition.
 - () Loss of proteins in urine.
 - () Blockage of lymphatic vessels.
 - () Heart failure.

BLOOD

Single-celled organisms obtain their needs for food and oxygen from their surrounding medium and also excrete waste products from food combustion into the surrounding fluid. However, the human body contains millions of cells that cannot access the external environment to obtain their needs for food and oxygen. Therefore, an intermediary between them and the external environment is necessary... this intermediary is blood.

Functions of Blood:

1. Carries oxygen from the lungs to the tissues and carbon dioxide from the tissues to the lungs.
2. Carries absorbed food from the small intestine to the tissues.
3. Carries waste products from the combustion of nutrients to the excretory organs.
4. Helps regulate body temperature.
5. Carries hormones from endocrine glands to the tissues.
6. Contains white blood cells and antibodies that help protect the body from diseases.
7. Blood is a red, crimson fluid in the arteries, tending to bluish in the veins. It is opaque due to the presence of many suspended cells.
8. Blood consists of a fluid known as plasma, in which cells known as blood cells are suspended.

There are three types:

- Red Blood Cells (Erythrocytes)
- White Blood Cells (Leukocytes)
- Blood Platelets (Thrombocytes)

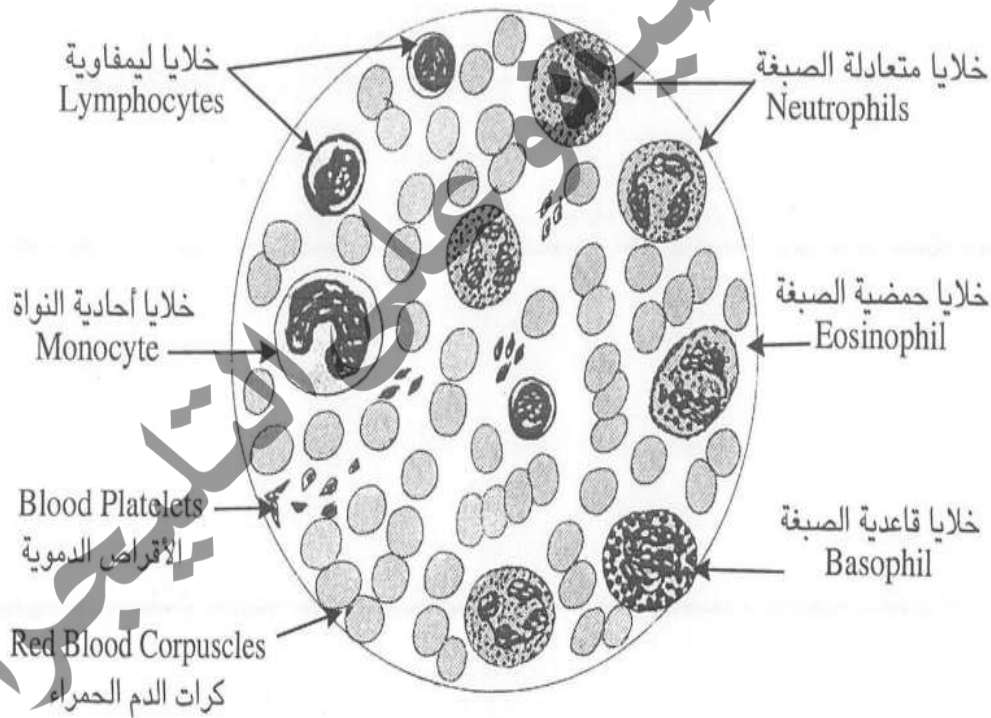
Erythrocyte Sedimentation Rate (ESR):

If blood clotting is prevented and the blood is placed in graduated tubes in a vertical position, the red blood cells are found to settle to the bottom, leaving the yellow fluid known as plasma. The sedimentation rate of red blood cells is 4 mm in the first hour and 7 mm in the second hour. It increases slightly during menstruation and pregnancy, but an increase to large amounts indicates the presence of disease within the body. It does not diagnose the type of disease but indicates its presence. The ESR increases in acute microbial diseases like tuberculosis, as well as in rheumatic fever, after surgeries and burns, and in cancer. It returns to normal levels after recovery from the disease.

Plasma:

A yellow fluid consisting of:

- Water 90%
- Inorganic salts 0.9% like sodium chloride, and salts of calcium, phosphorus, and a little potassium.
- Organic materials, which are divided into:
 - Plasma proteins 7.1%
 - Other organic materials 2.1%, including:
 - Nutritive substances on their way from the intestines to the tissues.
 - Waste products like urea on their way to excretory organs.
 - Hormones, enzymes, and antibodies.
 - Dissolved gases like oxygen, carbon dioxide, and nitrogen.



BLOOD FILM
(Microscopic Field)

Plasma Proteins:

The percentage of plasma proteins is 7 gm/100 ml. They are:

- Albumin
- Globulins
- Fibrinogen
- Prothrombin

Functions of Plasma Proteins:

1. Fibrinogen and one type of globulin (Prothrombin) are essential for blood clotting.
2. Proteins exist inside blood capillaries in a higher proportion than in tissue fluid, so they have an osmotic pressure that draws fluids from the tissues into the blood, maintaining a constant blood volume.
3. Give plasma viscosity that helps keep arterial blood pressure constant.
4. Keep the blood reaction (pH) constant.
5. Proteins carry antibodies that help defend the body.
6. Proteins carry some important substances for the body like fats, vitamins (A), iron, and iodine.
7. Combine with calcium salts to form the intercellular cement of the capillary walls, reducing their permeability.

Red Blood Cells (Erythrocytes):

Round discs without a nucleus, concave on both sides. Their wall is elastic, allowing them to change shape, enabling them to pass through capillaries often smaller in diameter than themselves.

The number of red blood cells in an adult male is 5 million per cubic millimeter of blood. It is higher in newborns than in adult males, but the excess cells break down in the first week after birth to reach 4.5 million.

Red blood cells live within the circulatory system for 120 days, after which they break down and are phagocytosed by the reticuloendothelial system, where hemoglobin is partially broken down into two parts:

- Protein (Globin): enters into protein metabolism within the body.
- Heme: the part that contains iron.

The body stores iron internally for use in manufacturing new red blood cells. The rest of the molecule is converted into yellow pigments (bilirubin) which the body disposes of via bile.

Manufacture of Red Blood Cells:

1. In the fetus in the first half of the pregnancy period, these cells are manufactured in the liver and spleen.

2. In the later months of pregnancy, the red bone marrow begins manufacturing red blood cells, and the liver and spleen stop producing them.

Factors Affecting Red Blood Cell Formation:

1. **Healthy Bone Marrow:** Red blood cells are manufactured after birth only in the bone marrow. If the bone marrow is exposed to X-rays or atomic radiation, its cells are destroyed and stop producing red blood cells.
2. **Decreased Oxygen Pressure in Arterial Blood:** A decrease in oxygen in arterial blood stimulates the production of red blood cells in the bone marrow. Their number can reach 8 million/mm³ in people living in very high mountainous areas where atmospheric pressure and consequently oxygen pressure are low.
3. **Nutrition:** The diet must contain:
 - (a) **Protein:** Necessary for the formation of hemoglobin. Protein of animal origin like liver and muscles has high nutritional value for hemoglobin production. An adult needs 1 gm/kg of body weight of protein.
 - (b) **Iron:** An essential element that enters into the composition of hemoglobin. The body needs very small amounts of iron daily, about 15 mg, because the body retains the iron from the hemoglobin of old cells to use in manufacturing new ones. The amount of iron needed increases in women to compensate for the amount of blood lost from the body during menstruation. Several factors determine the amount of iron absorbed by the body:
 1. Ferrous salts are easier to absorb than ferric salts.
 2. Inorganic iron is easier to absorb than organic iron.
 3. The presence of hydrochloric acid in the stomach facilitates iron absorption.
 4. The body's need for iron.
 - (c) **Copper and Cobalt:** Copper acts as a cofactor in the production of red blood cells. Cobalt enters into the composition of vitamin B12.
 - (d) **Vitamins:** Vitamin B12 is called the extrinsic factor for pernicious anemia. This factor combines in the stomach with an enzyme secreted by the gastric mucosa called the intrinsic factor for pernicious anemia, which is absorbed and stored in the liver. This factor is necessary for the complete maturation of red blood cells. Deficiency of vitamin B12 causes pernicious anemia, characterized by the circulation of immature red blood cells, accompanied by neurological disorders.
4. **Some Hormones,** most importantly Thyroxine: Thyroxine is necessary to stimulate the metabolism of all body tissues, including the bone marrow. Hormone deficiency causes a group of symptoms including anemia, while hormone excess does not lead to an increase in the number of red blood cells.

5. The Liver:

- Produces the protein that enters into the composition of hemoglobin.
- Stores iron.
- Also stores vitamin B12.

Types of Anemia:

- (1) Anemia due to the bone marrow's inability to produce blood cells, which may be caused by exposure of the bone marrow to X-rays or atomic radiation. In this type, the number of white blood cells and platelets produced in the bone marrow also decreases.
- (2) Nutritional Anemia: Results from protein deficiency in the diet.
- (3) Iron Deficiency Anemia: Results from iron deficiency in food.
- (4) Pernicious Anemia: Caused by vitamin B12 deficiency.

White Blood Cells (Leukocytes):

The number of white blood cells ranges from 8,000 to 10,000 per cubic millimeter. Their number increases in microbial diseases. White blood cells are produced in the bone marrow and in the lymphatic tissue.

Blood Platelets (Thrombocytes):

Yellow, disc-like bodies with a diameter about one-third that of red blood cells. Their number is about 250,000 per cubic millimeter. They aggregate when they encounter a rough surface. They are produced in the bone marrow. Platelets play an important role in blood clotting.

Blood Coagulation:

Blood is a fluid as long as it is inside healthy-walled blood vessels. However, if it is placed in a test tube, for example, it turns into a solid mass after minutes... this is known as a blood clot. If the clot is left for a while, it retracts and becomes suspended in a yellow fluid known as serum. If the clot is examined under a microscope, it is found to consist of many fibers (intersecting with each other forming a network), with blood cells trapped between these fibers.

Diseases Where Blood Does Not Clot:

1. **Hemophilia:** A hereditary disease that affects males and is not affected by females, but they carry it to their male sons. In this disease, the patient's blood does not clot, making them susceptible to bleeding from even minor injuries like bruises.
2. **Vitamin K Deficiency:** The liver needs vitamin K to produce prothrombin. If vitamin K is low, the amount of prothrombin in the plasma decreases. The amount of vitamin K in the body decreases under the following conditions:
 - Deficiency in the diet.
 - Killing the microbes present in the large intestine that produce it, after taking microbe-killing substances orally (like broad-spectrum antibiotics).

- Absorption of vitamin K from the intestines requires fat absorption. Its level in the blood decreases if fat absorption is impaired.
- In some conditions, liver cells are unable to produce prothrombin even in the presence of vitamin K. These conditions lead to death at an early age.

Hemorrhage:

Hemorrhage is the escape of blood from blood vessels. It is divided into two types:

(1) **External Hemorrhage:** Where blood exits the blood vessels to the outside of the body, such as bleeding from a sharp instrument injury, bleeding from the mouth, or postpartum hemorrhage.

(2) **Internal Hemorrhage:** Blood exits the blood vessels but does not appear outside the body; instead, it collects in a body cavity, such as bleeding into the peritoneal cavity or inside the cranial cavity. In this case, the occurrence of hemorrhage is indicated by the resulting symptoms.

(3) **Risks of Hemorrhage:** The danger of hemorrhage depends on the following factors:

1. The amount of blood the person loses.
2. The speed of blood loss.
3. The patient's health before the hemorrhage.

(4) **Symptoms of Hemorrhage:**

1. Pallor of the skin and mucous membranes.
2. Dry mouth and lips, and intense thirst.
3. Low blood pressure with increased rate and weak pulse.
4. Increased respiratory rate.
5. Ringing in the ears, visual disturbance, and possibly fainting occurs.

Note:

If the volume of lost blood exceeds $\frac{1}{3}$ of the blood volume in the circulation, the body is compensated with fluids by one of the following methods:

- a. Normal saline solution (0.9%) or a glucose solution, but these fluids do not have a significant effect as they quickly leave the circulation for the tissues.
- b. Plasma, which is the best substitute for the lost fluid.
- c. Blood transfusion.

Blood Transfusion:

Blood transfusion is used in the following cases:

- Cases of severe hemorrhage.
- Severe anemia if hemoglobin drops to 50%.
- Some blood diseases: hemophilia, hemolytic disease of the newborn, and blood cancers (leukemias).
- Shock, whether from burns or surgery.
- Carbon monoxide poisoning.

Precautions necessary before blood transfusion:

- Blood must be taken from a healthy volunteer, free from diseases transmitted through blood, such as viral hepatitis and AIDS.
- Sodium citrate or heparin is added to the blood to prevent it from clotting before injection.
- The blood must be of the same blood group as the patient to avoid complications for the patient.
- Blood is usually stored at a low temperature (below 4°C), and when used, it is warmed to about 37°C and transfused slowly.

Blood Groups:

Blood groups are divided based on the presence of protein substances on the red blood cell wall into groups. These protein substances are divided into two types, one called "A" and the other "B". Accordingly, blood is classified into group (A), (B), (AB) if it contains both types of protein. If the cells lack both types, the group is considered (O). For each of these proteins, there is a corresponding antibody in the plasma. For example, for protein (A), there is antibody (anti-A, alpha). Thus, if the protein meets its corresponding antibody, the red blood cells clump together (agglutinate), then their walls break down (hemolyze), and their contents are released outward, causing fatal complications. Naturally, therefore, a person whose cells contain protein (A) cannot have plasma containing antibody (anti-A, alpha); instead, it contains antibody (anti-B, beta).

Hence, we see that blood contains 4 groups or types:

(Table: Blood Groups)

Group Name	Protein on Cells	Antibody in Plasma
A	A	Beta (anti-B)
B	B	Alpha (anti-A)
AB	A and B	– (None)
O	– (None)	Alpha and Beta (anti-A & anti-B)

It is noted that the complications that occur in the body when transfusing blood from a group other than the patient's group occur in the transfused blood and not in the patient's blood, because the transfused red blood cells are few compared to the volume of the patient's plasma. Therefore, it is preferable to transfuse the patient with blood of the same group. If this is not found, blood of group (O-) is transfused because its cells do not contain any of the proteins (A), (B).

To know the patient's blood group, two drops of their blood are placed on a glass slide; one is mixed with serum containing antibody (anti-A, alpha) and the other with serum containing antibody (anti-B, beta). If the cells agglutinate with (alpha), then it is group (A). If they

agglutinate with (beta), it is group (B). If they agglutinate with both (alpha and beta), it is group (AB). If they do not agglutinate, it is group (O).

Symptoms of transfusing blood from a group other than the patient's:

1. The red blood cells clump together and block some blood capillaries, causing severe pain in the body and in the heart.
2. The pulse rate increases.
3. Shivering and chills occur.
4. The plasma bilirubin level rises as hemoglobin is released from the broken cells into the plasma.
5. The urine turns red like the color of plasma due to the presence of hemoglobin.
6. Kidney failure may occur, leading to death.

The Rh Factor:

This factor is named after the Rhesus monkey. It is found that 85% of humans have this factor on their blood cells (Rh positive), while the remaining 15% lack this factor on their blood cells (Rh negative). The importance of this factor appears in the following two cases:

1. If an Rh negative person receives a transfusion of Rh positive blood... antibodies against the Rh factor form in their blood. If they receive another Rh positive blood transfusion later, these antibodies will react with the transfused blood cells.
2. If an Rh negative woman marries an Rh positive man and becomes pregnant with an Rh positive fetus, antibodies may form in the mother's blood against the Rh factor. These antibodies cross the placenta to the fetus, causing hemolysis of the fetal red blood cells, leading to fetal death either in utero or after birth (hemolytic disease of the newborn).

REVIEW QUESTIONS: BLOOD

- Functions of blood.
- Types of plasma proteins and their functions.
- Precautions and complications of blood transfusion.

Complete the following:

a. The normal number of

red blood cells in an adult male is per cubic millimeter, and these cells live for days, and their formation is affected by factors including

b. Anemia is And Types of anemia include , And

C / The number of blood cells is They are made in and their function is

D / The number of platelets is They are made in and are necessary for the process of

E / The erythrocyte sedimentation rate (ESR) increases during such as

F / Blood does not clot in some cases, such as the disease or a deficiency of vitamin

G / Hemophilia is and its symptoms are

(3) Choose the correct answer from the following:

A) Pernicious anemia results from:

- Iron deficiency.
- Vitamin B12 deficiency.
- Lack of protein in the diet.
- Malignant disease of the bone marrow.

B) The erythrocyte sedimentation rate (ESR) increases in all of the following cases except:

- Pregnancy.
- Tuberculosis.
- Athletic effort.
- Rheumatic fever.

THE RESPIRATORY SYSTEM

Respiration is the process of exchanging different gases between the blood and the surrounding air through the respiratory system. Its purpose is to provide the blood with oxygen and rid it of carbon dioxide, in addition to regulating the temperature and acidity of the blood, among other constant functions.

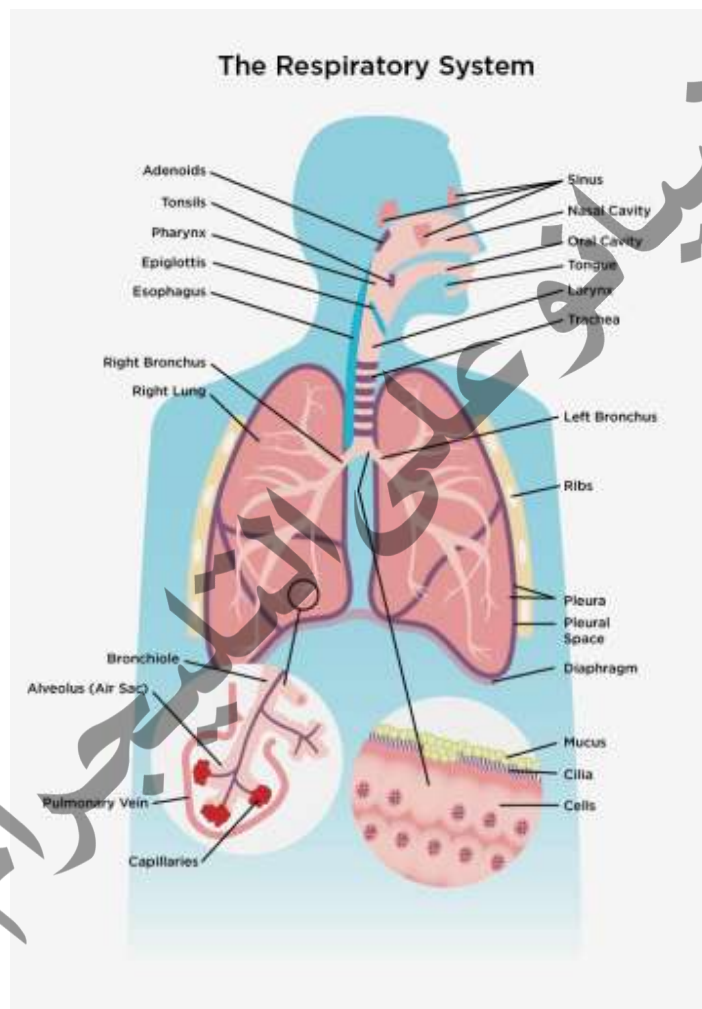
Respiration is divided into:

(1) **External Respiration:** Which is the exchange of gases between the blood and the air in the lungs.

(2) **Internal Respiration:** Which is the exchange of gases between the blood and the body's cells.

The mechanism of human respiration:

The air is drawn in through the nose, where it is warmed, moistened, and filtered from dust by the nasal conchae and hairs.



- **Mechanism of Inspiration:**

Inspiration is the expansion of the chest in all directions and the entry of air. The process of inspiration occurs as a result of the contraction of the diaphragm, which flattens it, thus increasing the vertical diameter of the chest. At the same time, the outer layer of intercostal muscles contracts, raising the ribs upwards and increasing the chest's diameter from the sides (anterior and posterior). Thus, during inspiration, the chest expands in all directions, the air pressure inside decreases, and air rushes in from the nose to the chest.

- **Mechanism of Expiration:**

Normal expiration is a passive process and is merely the relaxation from inspiration. The contraction of the diaphragm and intercostal muscles ceases, the chest volume returns to normal, pressure rises, and an amount of air equivalent to that which entered during inspiration is expelled. No muscles are involved in normal expiration. However, in cases of respiratory distress, expiratory muscles contract, including the inner layer of intercostal muscles.

Regulation of Respiration

The respiration process is regulated by a center located in the medulla oblongata known as the respiratory center. The rate and depth of breathing change in many situations according to the body's needs because the respiratory center is influenced by several factors, which are:

1. **Cerebral Cortex:**

Although breathing is an involuntary process, the muscles that perform it are voluntary. Therefore, we see that a person can stop their breathing for a period. Likewise, a person can increase the rate and depth of their breathing for a period.

2. **Hypothalamus:**

The center for regulating body temperature. We find that in hot weather, breathing becomes rapid and shallow, the purpose of which is to evaporate fluid from the respiratory system and lose heat, not for respiration. This happens as a result of the hypothalamus stimulating the regulatory center, thus increasing the respiratory rate. It also influences breathing during Reflex actions.

- **Reflex Actions:**

- Breathing stops during swallowing, defecation, and coughing.

- Reflex actions from the circulatory system.
 - (A) Respiratory rate decreases if arterial blood pressure rises.
 - (B) Respiratory rate increases if venous blood pressure increases.
- The most important influence on respiration is the inhibitory signals arriving via the vagus nerve to the inspiratory center.

Chemical Factors:

- (A) Increase in carbon dioxide in the arterial blood.
- (B) Increase in blood acidity.
- (C) Decrease in oxygen in the blood.

Hypoxia (Oxygen Deficiency)

Attributed to cases of oxygen deficiency in the cell. There are four types:

1. **Hypoxic Hypoxia:**

Results from a deficiency of oxygen reaching the blood from the lungs, as in:

- High altitudes.
- All lung diseases.
- Being in enclosed places.

2. **Anemic Hypoxia:**

This type occurs in the following conditions:

- Anemias of all types.
- Poisoning by carbon monoxide (CO), which is found in high concentrations in cooking gas. The affinity of hemoglobin for carbon monoxide is 210 times its affinity for oxygen.

3. **Stagnant (Circulatory) Hypoxia:**

This type results from a decrease in the amount of blood passing through the tissues. It occurs in:

- Low arterial pressure or high venous pressure, as in cases of heart failure.
- During exposure to cold, blood vessels constrict, reducing the amount of blood passing through tissues like the skin.

4. **Histotoxic Hypoxia:**

Occurs in cases that poison the oxidation and reduction enzymes inside the cell, such as poisoning with cyanide or alcohol.

REVIEW QUESTIONS ON RESPIRATION:

1. Explain the mechanism of:
A / The process of inspiration.
B / The process of expiration.
 2. What are the types and causes of hypoxia?
 3. Complete the following:
A / Respiration processes are regulated by a center located in and it is influenced by several factors, which are
B / Due to poisoning with carbon monoxide gas, hypoxia occurs, while poisoning with alcohol causes hypoxia.
 4. Choose the correct answer from the following:
(1) The respiratory rate increases in all of the following cases except:
☐ () Fever.
☐ () Swallowing.
☐ () Emotional states.
☐ () Oxygen deficiency.

(B) Stagnant hypoxia results from:
☐ () Anemia.
☐ () Poisoning with cyanide.
☐ () Heart diseases.
☐ () High altitudes.
-

DIGESTIVE SYSTEM

It is the system that mechanically and chemically breaks down complex food into simpler, absorbable substances that can pass through the cell membrane to be utilized by the body. This process is called "digestion", and the system responsible for it is the "digestive system".

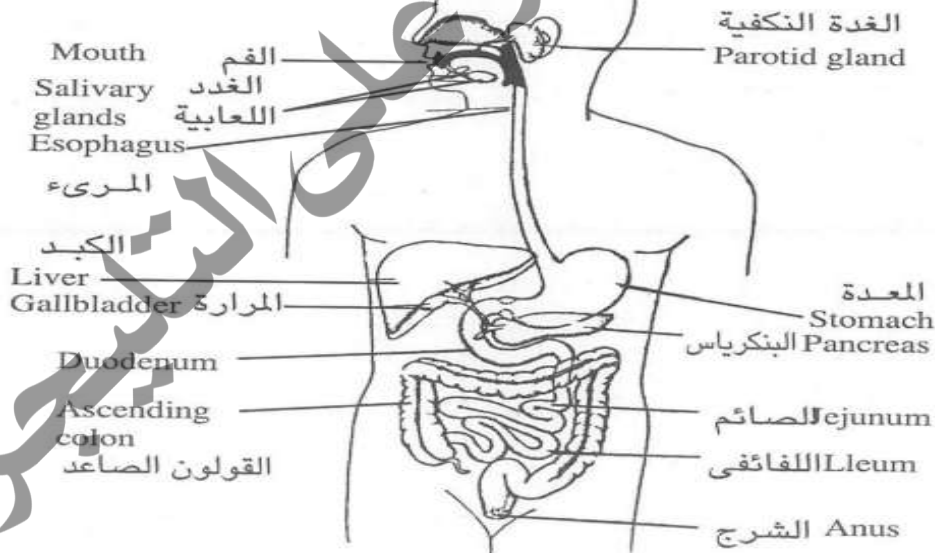
The digestive system consists of:

- (1) **The Alimentary Canal:** Which includes the mouth, esophagus, stomach, small intestine, and large intestine.
- (2) **Digestive Glands:** Such as the salivary glands, liver, and pancreas.

- The alimentary canal has a layered structure:

1. Mucosa
2. Submucosa
3. Circular muscle layer
4. Longitudinal muscle layer
5. Serosa (Peritoneum)

Peristalsis is the wave-like movement that propels food through the digestive system. It is an involuntary movement of the muscular walls, occurring when the canal is stretched. Digestion begins in the mouth.



The digested food goes into the blood in a process known as absorption. The remaining waste products of food that are not absorbed are expelled from the body through the anus in the feces.

- **Digestion in the Mouth:**

Digestion in the mouth is carried out by saliva. Saliva is secreted by 3 pairs of salivary glands:

- Parotid glands
- Submandibular glands
- Sublingual glands

There are also small glands scattered throughout the mucous membrane lining the mouth that secrete mucus.

- **Mechanism of Saliva Secretion:**

Salivary glands secrete saliva continuously, but secretion increases when eating food due to:

1. **Unconditioned reflex:** Resulting from stimulation of the oral mucosa by food, chewing gum, or a dentist's tools.
2. **Conditioned reflex:** Resulting from thinking about food, smelling it, or seeing it. The conditioned reflex requires habit to occur.

- **Functions of Saliva:**

1. Moistens food to facilitate swallowing.
2. Dissolves some solid foods so they can be tasted.
3. Cleanses the mouth and teeth of food remnants. In fever, when saliva secretion decreases, these remnants accumulate on the tongue, forming a white layer with a bad odor.
4. **Digestion:** Digests cooked starchy materials by the enzyme ptyalin; digests malt sugar by maltase.
5. Aids in speech because it moistens the soft parts of the mouth.
6. Protects the oral mucosa from the effect of acids and alkalis that may be present in food.
7. Dilutes the acidity of hydrochloric acid present in the stomach.
8. Regulates the amount of water in the body, as a person feels thirsty when the body's water content decreases.
9. Excretes some harmful substances from the body, such as lead, and some germs like the rabies microbe.

- **Digestion in the Stomach:**

The stomach is the widest part of the digestive system and is divided into three regions:

- **Fundus:** The dilated part above the cardiac orifice when a person is standing.
- **Body:** The dilated part that follows it.
- **Pylorus:** The part that follows the body and opens into the duodenum.

- **Gastric Juice:**

A highly acidic fluid, with a daily volume of 2-3 liters. It is secreted by tubular glands that open into the gastric mucosa. The glands in the body and fundus differ from those in the pylorus and around the cardiac orifice; the latter secrete only mucus, while the glands of the body and fundus secrete hydrochloric acid, enzymes, and mucus.

Functions of Gastric Juice:

(1) Hydrochloric Acid:

- Secretes the enzyme pepsin in an inactive form called pepsinogen. Hydrochloric acid activates pepsinogen and converts it into active pepsin, also providing the acidic medium necessary for its action.
- Helps dissolve some inorganic salts, such as calcium carbonate and iron salts, thus facilitating their absorption.
- Kills all germs that enter the body with food, except for the tuberculosis bacillus and Ascaris eggs.
- Hydrolyzes some disaccharides and fats.
- Coagulates the caseinogen in milk.
- When it reaches the duodenum, it stimulates a hormone called secretin. This hormone travels with the blood to the pancreas and liver, stimulating their secretion.

(B) Pepsin:

Pepsin is secreted in an inactive form that is activated by hydrochloric acid and only works in an acidic medium. Pepsin acts on proteins and converts them into proteoses and peptones.

(C) Function of Rennin (Chymosin):

Causes milk to coagulate in infants.

(D) Gastric Lipase:

The lipase enzyme digests fatty substances.

(E) Mucus:

Protects the mucous membrane lining the stomach from the effect of hydrochloric acid.

• Digestion in the Small Intestine:

Digestion in the small intestine is carried out by 3 important digestive juices that pour into it simultaneously, and their actions complement each other.

These juices are:

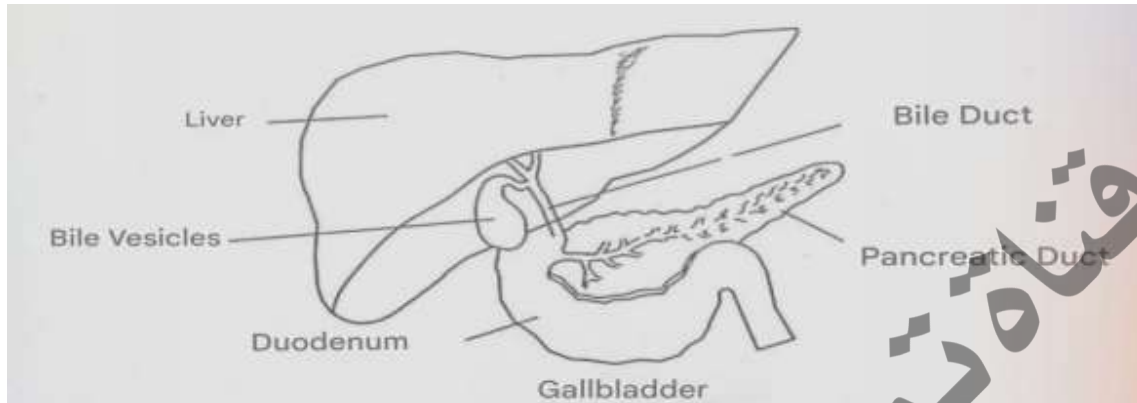
- Pancreatic juice.
- Bile.
- The intestinal juice itself (succus entericus).

Pancreatic Juice:

The pancreas is a gland in the upper abdominal cavity, under the stomach. It has two secretions: an internal secretion, which is the hormone insulin, and an external secretion, which is the pancreatic juice that passes via a duct to the duodenum.

Composition of Pancreatic Juice:

A highly alkaline fluid, with a daily volume of 750 cc. It consists of: 97.6% water, sodium bicarbonate (to neutralize hydrochloric acid), and enzymes for digesting all types of food.

**Liver Secretion "Bile":**

The liver is the largest gland in the body and is essential for life. It has several important functions, including the secretion of bile.

Bile:

A fluid that is alkaline upon secretion. It is secreted continuously but only reaches the duodenum after eating because it is stored in a sac located under the liver known as the gallbladder.

Functions of Bile:

1. Converts fatty materials into small particles called "emulsified fat," facilitating its digestion.
2. Helps in the absorption of fats and fat-soluble vitamins.
3. Stimulates the liver and increases its secretion.
4. Stimulates intestinal movement (peristalsis).
5. Protects the intestines from putrefaction.

Functions of the Gallbladder:

- (1) Stores bile until needed.
- (2) Concentrates bile, as the gallbladder volume is 50 cc, so the water in all the bile secreted by the liver is absorbed, reducing its volume and increasing its concentration.
- (3) Absorbs, along with water, sodium bicarbonate, making the bile's reaction acidic, which dissolves calcium salts and prevents the formation of gallstones.
- (4) Adds mucus and cholesterol salts to the bile.

The gallbladder contracts and bile reaches the duodenum in the following circumstances:

- A) After eating, due to a reflex action of the gallbladder via the vagus nerve, causing the wall muscles to contract and the sphincter muscle guarding the bile duct to relax, allowing bile to pass into the duodenum.
- B) The presence of food, especially fatty materials, in the duodenum stimulates the secretion of the hormone cholecystokinin, which travels with the blood to the gallbladder, causing it to empty into the duodenum.
- C) Some bile salts are absorbed with food and travel with the blood to the liver, stimulating its cells to secrete bile.

Functions of the Large Intestine:

1. **Absorption of water:** The large intestine absorbs most of the water that enters it, allowing no more than 100-150 cc of water to be lost in the feces.
2. **Digestion of cellulose:** The human digestive system lacks an enzyme to digest cellulose, as is the case in herbivorous animals. However, cellulose in the intestines is acted upon by bacteria.
3. **Absorption:** The large intestine can absorb a diluted treatment solution of glucose, amino acids, and salt. This property is used in therapy during coma.
4. **Excretion of some salts** such as calcium, iron, and mercury.
5. **Secretion of mucus:** This mucus helps bind the parts of the feces together and prevents the feces from rubbing against the intestinal mucosa.
6. **Storage of feces** until expulsion in the defecation process.

- **Nerves of the Digestive System:**

The digestive system is supplied by nerves from a system called the autonomic nervous system. This system regulates all involuntary movements in the body and is divided into two parts.

1. The sympathetic nervous system.
2. The parasympathetic nervous system.
3. **The Sympathetic Nervous System:**
Inhibitory... meaning it decreases movement in the digestive system.
4. **The Parasympathetic Nervous System:**
Stimulatory; it increases the intensity of contraction of the intestinal muscles and relaxes the sphincter muscles, moving food from one place to another, thus aiding digestion and absorption.
5. **Absorption:**
A vital process by which digested food is transferred to the blood. Most absorption occurs in the small intestine for the following reasons:
 1. It is the longest part of the digestive tract, reaching about 6.5 meters.
 2. It is rich in blood and lymphatic vessels.
 3. It contains villi.

6. **Villi:**

Finger-like projections that protrude into the intestinal lumen. They are covered internally by a mucous membrane and contain within them:

1. An artery or arteries that branch into a network of blood capillaries.
2. A small vein arising from the collection of blood capillaries.
3. A lymphatic vessel called a "lacteal".

The lymphatic vessels gather together and empty into a duct known as the thoracic duct, which carries lymph to the superior vena cava. The small veins gather and empty into the portal vein, which passes through the liver.

From this, we see there are two paths for absorbed food:

- a) Via the blood to the portal vein, then the liver.
- b) Via the lymph to the venous blood before it enters the heart.

QUESTIONS:

(1) What are:

- A) The functions of hydrochloric acid?
- B) The functions of the gallbladder?

(2) What happens if:

- A) Saliva secretion decreases?
- B) A stone forms in the bile duct?

(3) Complete:

- A) The digestive system is supplied by nerves, whose function is, and nerves help with
- B) Saliva secretion increases through the reflex, caused by, and the reflex.
- C) Absorption occurs in the small intestine because it is

(4) Choose the correct answer from the following:

A) Bile is responsible for all of the following except:

- (1) Absorption of fats.
- (2) Absorption of iron.
- (3) Stimulating the liver.
- (4) Stimulating intestinal movement.

B) The sympathetic nerves that supply the digestive system:

- (1) Decrease intestinal movement.
- (2) Increase intestinal movement.
- (3) Aid digestion.
- (4) Aid absorption.

METABOLISM

These are the chemical changes that occur in the food absorbed from the digestive system. The purpose of these changes is to supply the body with the necessary energy for its growth and to compensate for what has been consumed from its compounds.

The unit used to measure this energy is the large Calorie (kilocalorie).

- **Definition of a Large Calorie (Calorie):**

- It is the amount of heat required to raise or lower the temperature of one liter of water by one degree Celsius.
- The caloric value of nutrients when taken outside the body:
 - Each gram of carbohydrates gives 4.1 Calories.
 - Each gram of fats gives 9.3 Calories.
 - Each gram of proteins gives 4.1 Calories.

- **Metabolism of Proteins:**

- Proteins are the basic compounds of protoplasm. Proteins are absorbed from the digestive canal in the form of amino acids, which pass via the blood to the liver and then to the rest of the body. Amino acids, after absorption, have uses such as:
 - Building new proteins inside the body, i.e., replacing consumed protoplasm, also building new tissues in cases of growth in children, and forming fetal tissues and milk in nursing mothers.
 - Forming the secretions of endocrine glands.
 - Supplying the body with energy and heat.
- The combustion of protein materials inside the body produces waste products that the body usually disposes of through urine. These wastes are:
 - A / Urea
 - B / Ammonia
 - C / Uric acid
 - D / Creatinine
 - E / Sulfur.

- **Metabolism of Carbohydrates:**

- Carbohydrates are the main fuel for the body, providing more than 2/3 of its energy needs. Carbohydrates are absorbed from the small intestine in the form of glucose, which is carried by the blood to the liver and then to the rest of the body.
- The blood glucose level before eating ranges from 80 mg% to 120 mg/100 cc. It is lower in venous blood than arterial blood. The blood sugar level rises during absorption, reaching a maximum of 120-160 mg%, then returns to its original level within 3 hours of eating.
- **Uses of Glucose inside the Body:**
 - (1) Glucose goes to the liver where it is stored as glycogen. Liver glycogen is considered a general store of carbohydrates for the body. If the blood sugar level drops, liver glycogen is converted back to sugar. Note that liver glycogen differs from muscle glycogen, which cannot be used by any tissue other than muscles. The muscle uses it during contraction, and it cannot be converted back into blood glucose.
 - (2) Glucose is burned inside the body's cells into carbon dioxide, water, and thermal energy.
 - (3) Excess carbohydrates beyond the body's needs are converted into fat and stored in the body.
 - (4) It enters the composition of some compounds, like milk in lactating women and mucous secretions.
 - (5) **Excretion in urine:** Sugar does not usually appear in the urine unless its level in the blood exceeds 180 mg per 100 cc of blood. This is called the renal threshold for glucose.
- **Regulation of Blood Glucose Level:**

The blood glucose level depends on several factors, the most important of which are:

 - (1) **Stomach:**

We mentioned the regulation of the stomach. If a person ingests large amounts of glucose at once, it leads to vomiting. Also, the hormone enterogastrone is secreted, which reduces stomach movement and secretion.
 - (2) **Intestines:**

The rate of glucose absorption from the intestines is constant and does not depend on the amount of sugar in the intestines.
 - (3) **Liver:**

Absorbed glucose passes to the liver, which stores part of it as glycogen, preventing high blood

sugar. If the blood sugar level drops, liver glycogen is converted back to sugar to raise the blood sugar level again.

(4) The blood glucose level also depends on a balance in the action of several hormones. These hormones are:

A/ Pancreas:

The pancreas secretes two hormones (Insulin and Glucagon).

text

- **Effect of Insulin:**

Insulin lowers the blood sugar level as it:

- Helps convert glucose to glycogen and store it in the liver.
- Helps burn sugar in the tissues into carbon dioxide, water, and thermal energy.
- Helps store glucose as fat.
- Prevents the conversion of proteins into sugary materials.

- **Effect of Glucagon:**

Opposes the action of insulin, thus increasing blood sugar levels.

****B/ Anterior Pituitary Gland:****

Secretes a hormone known as the Diabetogenic Hormone, whose effect completely opposes insulin as it:

- Converts liver glycogen to glucose.
- Helps convert protein to sugar.
- Prevents the conversion of glucose in tissues to carbon dioxide.

****C/ Adrenal Gland:****

Consists of the cortex and medulla:

- ****Adrenal Cortex:**** Its hormones help convert protein to sugar, raising blood glucose.
- ****Adrenal Medulla:**** Secretes adrenaline, which helps convert liver glycogen to glucose.

D/ Thyroid Gland:

Secretes thyroxine, which stimulates metabolism in tissues and also helps convert glycogen to glucose.

From this, we see that the blood sugar level depends on the balance between the effect of insulin on one hand and the effects of several other hormones on the other.

(5) Kidneys:

If the blood glucose level exceeds the renal threshold for glucose (180 mg%), sugar appears in the urine.

- **Glycosuria:**

The appearance of sugar in the urine. This occurs as a result of:

1. An increase in blood glucose level.
2. A decrease in the renal threshold for glucose below normal.

Types of Glycosuria:

(1) Glycosuria not accompanied by high blood sugar:

- Renal Glycosuria: When the renal threshold for glucose is low.

(2) Glycosuria accompanied by high blood sugar:

- Alimentary Glycosuria: If a person consumes large amounts of carbohydrates at once, blood sugar rises significantly and sugar appears in the urine, but it returns to normal within 2-3 hours.

- Emotional Glycosuria: The blood sugar level increases in emotional states due to the secretion of adrenaline from the adrenal medulla, which converts liver glycogen to glucose.

- Pancreatic Glycosuria: Diabetes (insulin deficiency).

- Increased secretion of the following endocrine glands:

- Glucagon from the pancreas.
- Thyroxine from the thyroid gland.
- Diabetogenic hormone from the anterior pituitary gland.
- Hormones from the adrenal cortex.

- **Metabolism of Fats:**

Fats enter into the composition of cell protoplasm and are also stored in the body as fat droplets

within adipose connective tissue. Fats are digested in the body into fatty acids and glycerol. Fats are used in the body for the following purposes:

1. Converted into fats that enter the composition of body cells.
2. Burned into carbon dioxide, water, and energy. Fatty materials are an important source of heat inside the body.
3. A portion of fats is converted into carbohydrates.
4. Formation of milk fat in lactating women.
5. The excess is stored as fat droplets in adipose tissue. A person with a constant weight does not add any amount of fat to their stores; rather, all consumed fat is burned to supply the body with the necessary energy.

QUESTIONS:

1. What is meant by each of the following:
 - A) Metabolism
 - B) Large Calorie (Kilocalorie)
 - C) Glycosuria

2. What are the factors responsible for regulating blood sugar levels?

3. Complete the following:
 - A) The waste products resulting from the metabolism of proteins in the diet are
 - B) Glucose has many uses inside the body, including
 - C) Fats are used in the body for

4. Choose the correct answer from the following:
 - A) Sugar appears in the urine if its level in the blood exceeds:
 - () 80 mg/100 cc blood
 - () 120 mg/100 cc blood
 - () 180 mg/100 cc blood
 - () 300 mg/100 cc blood
 - B) The following hormones affect blood sugar levels except:
 - () Insulin.
 - () Adrenaline.
 - () Thyroxine.
 - () Parathormone.

REGULATION OF BODY TEMPERATURE

The normal human body temperature is 37°C and is almost constant. It may increase during physical exertion or digestion and may decrease during hunger or sleep. If a person works during the day and rests at night, this change does not exceed 0.5°C .

The temperature of organs inside the body varies according to the work performed by the organ. For example, skin temperature is lower than blood temperature (37°C), which is lower than intestinal temperature, and liver temperature is higher than any other organ.

- **Center for Regulating Body Temperature:**

The center for regulating body temperature is located in the hypothalamus. The hypothalamus is stimulated in one of two ways:

- Any change in the temperature of the blood passing through it.
- The skin contains nerve endings sensitive to heat and cold. These endings send signals to the hypothalamus to alert it to the temperature of the surrounding environment.

- **Changes that occur in the body when exposed to a hot environment:**

When exposed to hot weather, the hypothalamus stimulates the body's heat loss mechanisms. These mechanisms are:

- The skin.
 - The lungs.
 - The urinary and digestive systems.
 - Raising the temperature of ingested food to body temperature
-

The Skin:

The most important organ for heat loss. The body loses $7/8$ of its heat through the skin via the following methods:

Conduction:

When touching a cold object, the body loses heat.

- **Convection:**

If the air surrounding our body is heated, its density decreases and it rises to higher layers of the atmosphere, replaced by cold air.

- **Radiation:**

This is the loss of heat from a hot body to a colder one without heating the medium between them. This is how sunlight reaches the earth or how warmth comes from a heater.

- **Evaporation:**

The skin secretes sweat, which evaporates from the skin's surface. This evaporation occurs year-round. In winter, it gives the skin softness, while in summer it helps lose heat from the skin.

Sweat is of two types:

1. **Visible sweat:**

Observed during summer or during physical exertion. This type is not very effective in regulating body temperature because it does not evaporate from the body's surface.

2. **Invisible sweat (Insensible perspiration):**

Evaporates from the body's surface continuously, secreted both summer and winter, but its secretion increases in summer.

3. **The Lungs:**

The body loses about 5% of excess heat through the lungs by the following methods:

- Evaporation of water.
- Warming the inspired air and raising it to body temperature.

Changes that occur in the body when exposed to a cold environment:

When exposed to cold weather, the difference between skin temperature and ambient temperature increases. This leads to an increase in the amount of heat lost from the body through the skin. The body works to reduce the amount of heat lost and to increase the amount of heat produced inside the body. The blood vessels in the skin constrict (perhaps the skin temperature depends on the amount of blood within it). Therefore, we find that skin temperature drops, thus reducing the difference between skin temperature and ambient temperature, so less heat is lost through the skin.

- The amount of heat produced inside the body increases by the following methods:
 - Muscle contraction, either through physical effort or by muscle shivering.
 - The hypothalamus stimulates the secretion of thyroxine and adrenaline. These hormones help increase the body's metabolism and increase the amount of heat produced inside the body.

QUESTIONS:

1. What happens if:
 - A) The body is exposed to a hot environment?
 - B) The body is exposed to a cold environment?

2. Complete the following:

The center for regulating body temperature is located in and can be stimulated by

3. Choose the correct answer from the following:
 - A) If the body is exposed to a hot environment:
 - (1) Blood vessels in the skin constrict.
 - (2) The rate of sweat secretion increases.
 - (3) The metabolic rate increases.
 - (4) Muscle shivering occurs.

EXCRETION OF WASTE MATERIALS

We have previously explained how the body gets rid of carbon dioxide through the lungs, and how the digestive system gets rid of undigested food waste through the anus. It remains for us to talk about the system that rids us of excess water beyond the body's needs and also expels harmful substances present in the blood in a dissolved state... This system is the urinary system.

1- Urinary System

The important function of this system is to filter urine from the blood and expel it to the outside. The urinary system consists of the kidneys, ureters, bladder, and then the urethra.

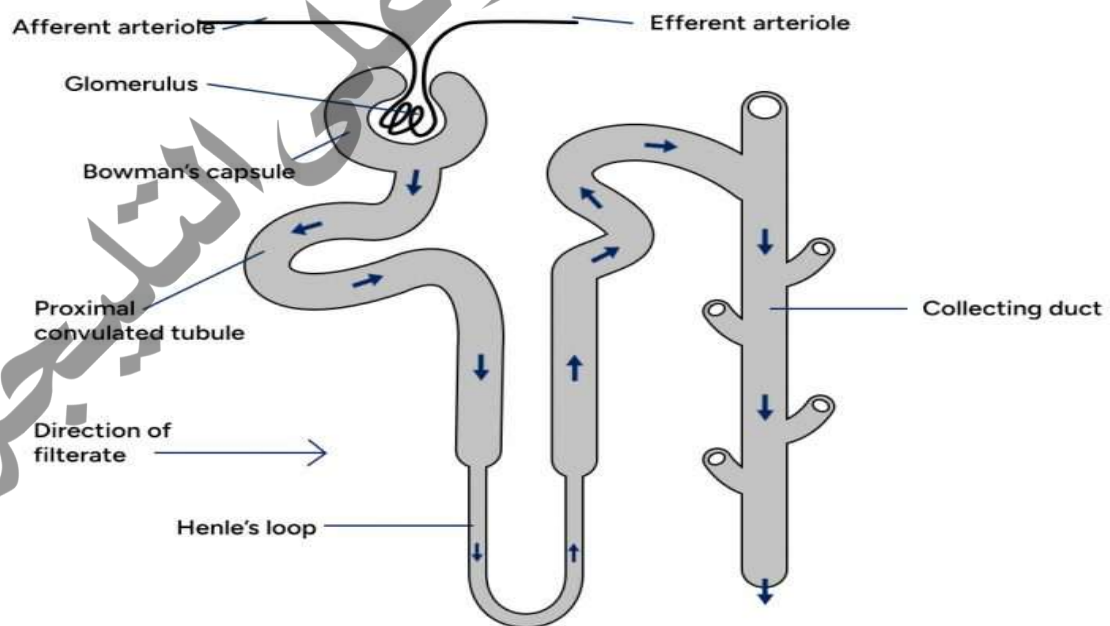
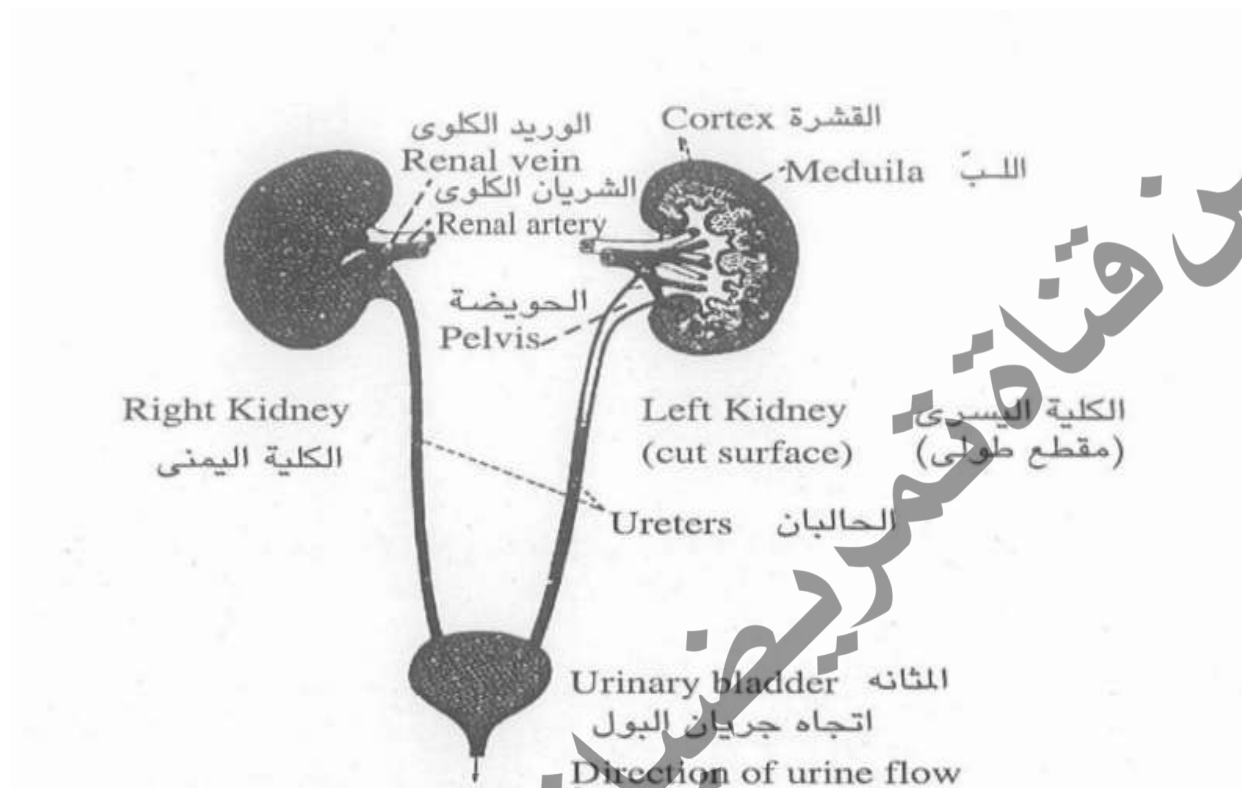
Functions of the Kidney:

- (1) Maintains the constant composition of the internal environment of the body.
- (2) Regulates the amount of fluids in the body. If the body's internal fluids increase, the volume of urine increases.
- (3) Regulates the osmotic pressure of the blood.
- (4) Maintains a constant blood pH by adjusting urine pH and secreting ammonia, which neutralizes acids produced inside the body.
- (5) Disposes of harmful wastes like urea.
- (6) Neutralizes some toxic substances that enter the body, such as benzoic acid.
- (7) Participates in maintaining blood pressure.

How the Kidney and its Unit Function:

The kidney is composed of a very large number of microscopic tubes called "nephrons" (renal units). Each unit begins with a spherical expansion into which a small branch of the renal artery enters and branches into fine blood capillaries. This expansion with its capillaries is called the "Malpighian corpuscle" or "glomerulus". Following this expansion is a long, coiled tube called the "renal tubule".

General Diagram of the Urinary System



The Malpighian corpuscle (glomerulus) acts as a filter, filtering from the blood as it passes through the capillaries a fluid resembling plasma in everything except that it does not contain plasma proteins. The filtrate flows into the renal tubule, where water and many useful dissolved substances, such as glucose and mineral salts, are reabsorbed and returned to the blood. At the same time, the renal tubules secrete substances harmful to the body and those in excess of its needs, adding them to the filtered fluid. The fluid formed as a result of these three processes (filtration, reabsorption, and secretion) is called "urine". The urine flows through the tubules until it reaches a collecting tubule. A number of these tubules join to empty into the "renal pelvis".

Urine

Urine is a pale yellow, acidic fluid with a distinct and characteristic odor. It contains useful salts such as sodium chloride, sulfate, and phosphate, as well as salts of urea, uric acid, and coloring materials.

The amount of urine is usually between 1 and 1.5 liters every twenty-four hours. This amount depends on the amount of fluids a person consumes, the amount of salts contained in the food, the ambient temperature, and the person's muscular activity.

- **Abnormal Urine:**

In special circumstances or pathological conditions, abnormal elements appear in the urine that would not normally pass through the kidney tissue. The most important abnormal elements in urine are:

- (1) **Albumin (Proteinuria):**

Its appearance in the urine indicates a pathological condition of the kidney, as the renal corpuscles have become unable to perform their function, allowing plasma proteins to pass into the urine.

- (2) **Sugar (Glycosuria)**

- (3) **Salt Deposits:**

These vary according to the reaction of the urine. If the urine is acidic, we find salts of uric acid and calcium oxalate. If the urine is alkaline, we find calcium phosphate.

- (4) **Casts:**

These are proteinaceous materials that managed to pass from the renal corpuscles and took the shape of the renal tubules they passed through. Casts can be bloody or fatty, depending on the tissue they are composed of.

- (5) **Blood Cells:**

White or red blood cells. If present in large numbers, they indicate an advanced pathological condition in the kidneys.

- (6) **Pus:**

This consists of white blood cells that have fought against various germs. It is evidence of inflammation in any part of the urinary system.

in diabetes, or renal casts in kidney disease, or bile pigments in liver disease.

- **The Process of Urination:**
Urine reaches the bladder from the ureters at a rate of about 2 cc per minute. This urine is stored in the bladder until the pressure inside it increases to a certain level, at which point the person feels the urge to urinate.
During urination, the muscles of the bladder wall contract and the sphincter muscle relaxes, allowing urine to exit through the urethra to the outside of the body.
The frequency of urination is affected by several natural factors; it increases if the fluids a person drinks increase and decreases if they decrease. It also decreases in the summer because the body loses a lot of water through sweat in hot weather.
The frequency of urination increases in some diseases like diabetes and prostate issues, and decreases in cases of kidney failure.

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QUESTIONS:

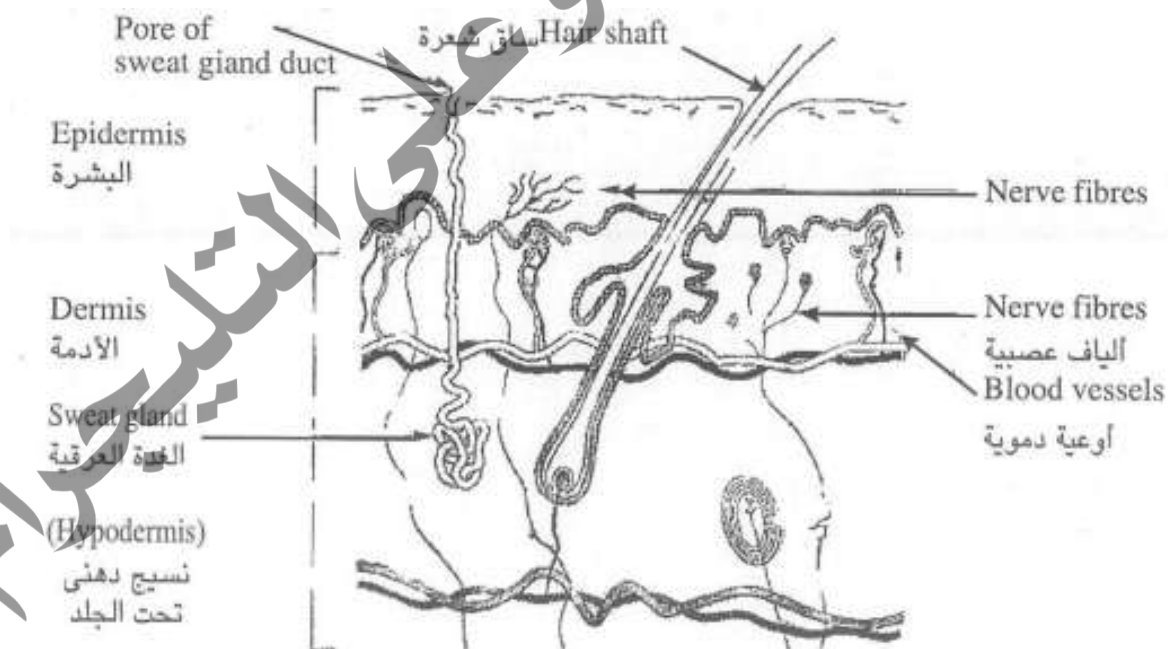
1. What are the functions of the kidney?
2. Label the parts of the nephron with a diagram.
3. Explain how the kidney performs its function of "urine formation".
4. Mention the most important abnormal elements that may appear in urine.
5. Complete the following:
A- The appearance of albumin in the urine indicates, while the appearance of sugar indicates
6. Choose the correct answer from the following:
A) The following elements in urine are considered abnormal except:
☐ Acetone.
☐ Albumin.
☐ Bile pigments.
☐ Sodium salts.

THE SKIN

The skin can be considered an important excretory organ for getting rid of water, inorganic salts, and a small amount of urea. This is done through the secretion of a fluid called "sweat". In the depth of the skin, there are clusters of cells called sweat glands, which have ducts that open onto the skin's epidermis. Their openings appear as small pores called skin pores. The amount of water lost through the skin ranges between 500-800 cc per day. After this fluid exits, it evaporates quickly. If its rate of evaporation equals its rate of production, the sweat is not seen and becomes imperceptible. However, if sweat is produced faster than it evaporates, it accumulates on the skin's surface as water droplets. The most important factors controlling the amount of sweat the body produces are ambient temperature and the amount of muscular activity the body performs. Sweat glands are under the control of the sympathetic (autonomic) nervous system. If the ambient temperature rises, or if blood temperature increases due to muscular effort or in some emotional states (fear, embarrassment), the sympathetic system is stimulated, affecting the sweat glands and causing sweat secretion.

The function of the skin is not limited to excretion; it has other important functions. The skin is a protective organ, safeguarding the tissues beneath it and acting as a cover that envelops the entire body. Its outer layer called the epidermis is composed of strong cells that protect the tissues from any mechanical, chemical, or microbial injury. The skin contains sensitive nerve endings responsible for distinguishing between heat and cold, pain, and touch.

In addition to excreting sweat, the skin secretes an oily substance from sebaceous glands present in the skin. We have previously discussed the important function of the skin in regulating body temperature.



Body Fluids

The human body contains a large percentage of water, amounting to about 60-70% of body weight. The percentage of water is lower in women and severely obese individuals and is high in newborns, then decreases with increasing age.

The water present in the human body is not in the form we drink it; rather, it enters into the composition of cells and the various body fluids.

- **Distribution of Water in the Human Body:**

- (1) **Intracellular Fluid (ICF):**

- This constitutes the largest part of the body's water. This water enters into the composition of cell protoplasm and contains potassium salts, which are the most important salts inside cells, as well as phosphate salts. Sodium salts are present in very small amounts inside cells, except for bone cells, which contain a high percentage of them.

- (2) **Extracellular Fluid (ECF):**

- Divided into two parts:

- A- Water inside blood vessels (in plasma).

- B- Water between cells (in interstitial fluid).

The most important salts present in the extracellular fluid are sodium, chloride, and bicarbonate salts, while potassium salts are present in very small amounts.

The composition of plasma and interstitial fluid is approximately similar, except that plasma contains albuminous materials (proteins).

Plasma circulates in the bloodstream, carrying nutrients. Exchange of constituent materials occurs between it and the interstitial fluid through the walls of the capillary blood vessels. Likewise, exchange of materials occurs between the interstitial fluid and the intracellular fluid through the cell membrane.

- **Water Balance in the Body:**

The body maintains a nearly constant amount of water thanks to the balance between the amount of water entering the body and the amount lost by the body.

The kidneys play an important role in maintaining this balance by controlling the amount of urine they secrete.

Amount of Water Entering or Gained by the Body

1. The body gains about 200 cc of water daily that is produced from metabolic processes. For example, when carbohydrate materials are burned, they turn into carbon dioxide, water, and energy.
2. The solid food consumed by an adult contains about 300 cc of water. For example, bread, meat, and fruit contain 25-75% of their weight as water.
3. A person drinks about 1000 cc of water daily. The more sweat is produced, as in summer, the more a person needs to drink larger quantities of water.

Adding the previous quantities together, we find that the body gains about 1500 cc of water daily

Amount of Water Lost by the Body

First: Water Lost Through Urine:

The amount of urine in an adult is estimated at about 1500 cc daily and should not be less than 1000 cc in order to dispose of metabolic waste products. The smaller the urine volume, the higher the concentration of substances in it and the higher its specific gravity. The kidneys control the volume of urine to keep the body's fluid composition constant.

Second: Water Lost Through Ways Other Than Urine:

(1) Water Lost Insensibly by the Body:

A / Water vapor present in exhaled air. Its amount depends on the rate and depth of breathing.
B / Water vapor lost through evaporation from the skin. Its amount depends on temperature, air speed, humidity, and the person's clothing. It is very important in regulating body temperature and this water is not related to sweat.

(2) Sweat:

This is the secretion of the sweat glands present in the skin. Its main functions are regulating body temperature and eliminating excess water, harmful substances, and salts. The amount of sweat varies from person to person and in the same person depending on weather conditions and the activity they are doing.

In addition to water, sweat contains amounts of sodium and potassium salts. If a person loses a large amount of sweat and then replaces it by drinking water only, a salt deficiency occurs. Therefore, one should drink ideal water containing an amount of table salt.

(3) Water Lost in Feces:

The amount of water lost in feces varies according to the nature of the intestines; it increases in intestinal catarrh and diarrhea and decreases if constipation occurs.

On average, a person loses about 200 cc of water in feces daily.

- **Regulation of Water Amount in the Human Body:**

The kidneys play an active role in regulating the amount of water in the body, while the lungs and skin have a passive role. The amount of urine a person produces is controlled by some endocrine glands, such as the pituitary gland. The amount of urine decreases if sweat secretion increases, as in hot weather, and increases if a person consumes a large amount of fluids or water.

In some pathological conditions, a person loses an additional amount of fluids through ways other than those mentioned. For example, in cases of burns, the victim loses a large amount of body fluids through the damaged skin surface. Likewise, if a person suffers from vomiting or diarrhea, they lose an amount of fluids that must be accounted for when balancing body fluids.

Calculating Fluid Balance for a Patient:

In pathological conditions and after surgical operations, a record called a "fluid balance" chart must be made for the patient. This chart records the fluids lost by the body and the fluids given to the patient, whether orally or by injection. There must be a balance between the fluids given to

the patient and the fluids lost, so that the amount of fluids in the body does not decrease or increase, as both cases are harmful and must be avoided.

When calculating the amount of fluids lost by the body, we must take into account the water lost insensibly in exhaled air and through the skin, and the water lost in urine, feces, and sweat, in addition to any abnormally lost fluids such as vomit, diarrhea, or from the skin surface in burns. The patient is then given an amount of fluids that balances the amount lost, either orally or by injection. It is preferable to administer fluids intravenously. These fluids should also contain an amount of salts equal to the amount of salts lost by the body.

QUESTIONS:

- (1) Draw and label the structure of the skin.
- (2) Mention the functions of the skin.
- (3) Discuss the distribution of water in the human body.
- (4) What are:
A / The means by which the body gains water.
B / The means by which the body loses water.
- (5) Explain how to calculate fluid balance for a patient.
- (6) Complete the following:
A) Among the means by which the body gains water are
B) Among the means by which the body loses water are
C) The amount of urine in an adult is estimated at about and should not be less than in order to get rid of
- (7) Choose the correct answer from the following:
A) Water constitutes about of an adult human's body weight.
 - (1) 40%
 - (2) 60%
 - (3) 10%
 - (4) 85%

ENDOCRINE GLANDS

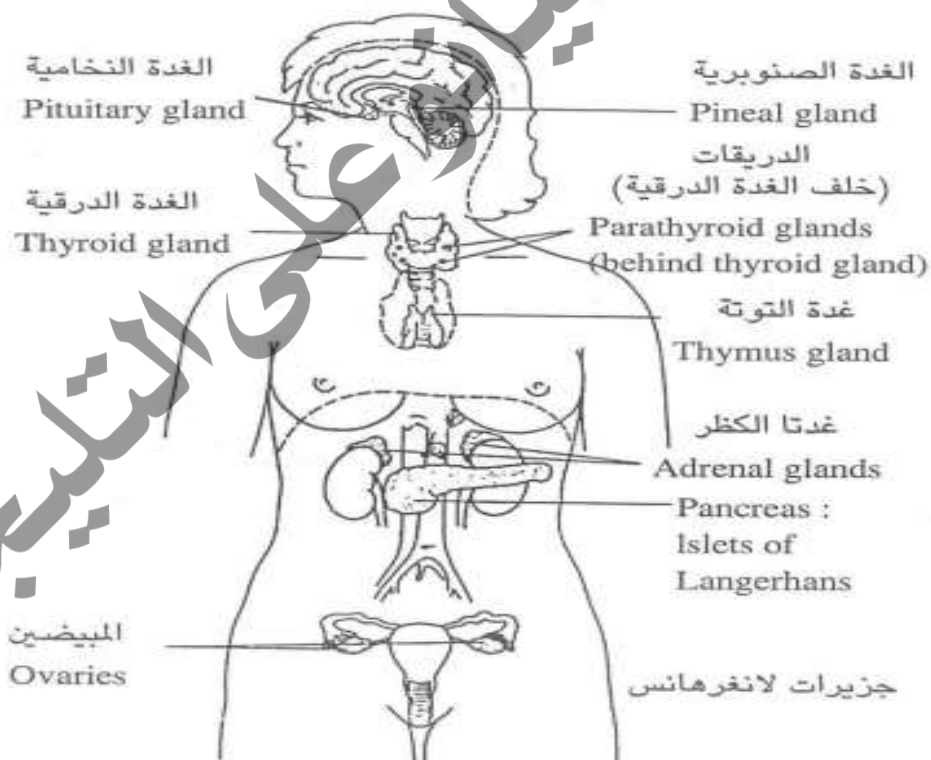
- **Glands in the body are of two types:**

(1) Glands that have a duct carrying their secretion to a cavity of the body, such as salivary glands, known as exocrine glands.

(2) Glands that do not have a duct to carry their secretion; instead, the secretion passes into the blood, which carries it to the tissues inside the body. These are known as glands of internal secretion or endocrine glands. The secretion of endocrine glands is known as a hormone.

- **There are a very large number of endocrine glands in the body, the most important of which are:**

- Pituitary gland
- Thyroid gland
- Parathyroid glands
- Adrenal glands
- Pancreas
- Sex glands (Gonads)



Anatomical Locations of the Main Endocrine Glands in the Body

Pituitary Gland

A small gland weighing between 0.5-0.8 grams, located in a cavity within the skull known by its name (sella turcica).

The pituitary gland is divided into two parts "lobes":

- An **anterior lobe** that secretes several hormones and is known as the glandular lobe.
- A **posterior lobe** connected to the hypothalamus by a neural stalk and known as the neural lobe.

Hormones of the Anterior Pituitary Lobe:

The anterior lobe of the pituitary gland is considered the master gland of the body's other endocrine glands, as it secretes hormones that regulate the function of those glands. The hormones of the anterior pituitary lobe are:

(1) **Growth Hormone (Somatotropin):**

Stimulates the growth of all body tissues, including bones. It is secreted abundantly in children and during pregnancy. It also affects the metabolism of carbohydrates and opposes the action of insulin. If the secretion of this hormone increases before puberty, the person becomes a giant (gigantism). If its secretion is deficient before puberty, the person becomes a dwarf.

(2) **Thyroid-Stimulating Hormone (TSH)**

(3) **Adrenocorticotrophic Hormone (ACTH)** (Stimulates the adrenal cortex)

(4) **Gonadotropic Hormones** (Stimulate the sex glands)

(5) **Prolactin** (Lactogenic hormone)

Posterior Lobe of the Pituitary Gland:

The posterior lobe secretes two hormones.

A / **Antidiuretic Hormone (ADH, Vasopressin)**, which also works to constrict blood vessels.

B / **Oxytocin**, which stimulates uterine contractions.

Thyroid Gland

The thyroid gland is located in the neck in front of the larynx. It secretes a hormone known as thyroxine. Thyroxine contains iodine in its molecule (about 65% iodine by weight).

Functions of Thyroxine:

1. Stimulates physical, mental, and sexual growth.
2. Stimulates the body's metabolism.
3. Helps maintain a constant body temperature.
4. The thyroid gland secretes as a result of its stimulation by the Thyroid-Stimulating Hormone (TSH) from the anterior pituitary lobe.
5. Deficiency of thyroxine hormone secretion in childhood causes delayed physical, mental, and sexual growth, and the person becomes short and mentally deficient (cretinism).
6. Deficiency of the hormone after puberty leads to the disease "myxedema," characterized by a decrease in various body activities.
7. Increased secretion of thyroxine hormone (Exophthalmic Goiter) causes an increase in various body activities and may be accompanied by protrusion of the eyes (exophthalmos).

- **Simple Goiter:**

Iodine constitutes a large part of thyroxine (about 65%). If iodine is deficient in the diet, the gland enlarges and the number of its cells increases, but symptoms of its deficient secretion appear on the body. Goiter resulting from iodine deficiency occurs endemically in areas far from the sea (as the sea is the source of iodine in food), such as in oases. Iodine is currently added to food to prevent the disease.

The body's susceptibility to simple goiter increases in conditions that call for increased hormone secretion from the thyroid gland, such as puberty and during pregnancy. The gland size increases slightly, known as physiological goiter. The enlargement disappears when the cause ends. The gland size also increases when affected by diseases like "tuberculosis."

Also, the presence of some food substances helps enlarge the gland because it prevents iodine from entering it from the blood. These substances are cabbage, turnips, and soybeans.

- **Parathyroid Glands:**

Four small, round bodies adherent to the posterior surface of the thyroid gland, two on each side. The parathyroid glands secrete a hormone called "parathormone." This hormone regulates the level of calcium and phosphate ions in the blood.

Consequently, this hormone has a great relationship with bone hardness and neuromuscular conduction.

- **Function of Parathormone:**

(1) On the Kidneys:

If the phosphate level in the blood increases, the hormone helps secrete the excess in the urine to keep the calcium level constant.

(2) On the Intestines:

Increases the rate of calcium absorption.

(3) On the Bones:

If calcium is low in the blood, the hormone mobilizes calcium from the bones to compensate for the deficiency in the blood.

- **Symptoms of Increased Secretion (Hyperparathyroidism):**

- Increases the calcium level in the blood as a result of drawing it from the bones, making them soft, easily bent, brittle, and prone to fracture.
- Increased calcium ions in the blood lead to their deposition in tissues like muscles and their deposition in urine, giving stones.
- Decreased sensitivity of nerve conduction and weakness in all types of muscles.

- **Symptoms of Decreased Secretion (Hypoparathyroidism):**

Decreased secretion leads to increased activity of the neuromuscular system, causing contractions (tetany) in some muscles of the body. If these contractions affect the respiratory muscles, it can lead to death. This is known as "tetany" disease.

Tetany is not specific to parathyroid deficiency; it occurs in any condition where the calcium ions in the blood are deficient, such as:

- Deficiency of parathyroid gland secretion.
- Rickets.

- Diseases where fat absorption from the intestines is impaired, thus impairing the absorption of vitamin "D," which helps absorb calcium.
- Chronic inflammation of the kidney.

- **Pancreas:**

A gland in the upper abdominal cavity under the stomach. The pancreas consists of two glands:

(1) **An exocrine gland:**

This is the digestive juice of the pancreas, which is carried by a duct to the duodenum.

(2) **An endocrine gland:**

This consists of groups of cells known as the Islets of Langerhans. These islets contain two types of cells:

- **Alpha cells:** Secrete a hormone called glucagon.
- **Beta cells:** Secrete the hormone insulin.

- **Glucagon:**

Its secretion is stimulated by low blood glucose. Its function is to convert glycogen (animal starch) stored in the liver into blood glucose.

- **Insulin:**

An increase in blood glucose levels stimulates the secretion of insulin.

Functions of Insulin:

1. Helps store sugar in the liver as animal starch "glycogen".
2. Helps burn glucose in the tissues into carbon dioxide and water.
3. Prevents the conversion of proteins into starchy materials.

- **Symptoms of Insulin Deficiency:**

Leads to diabetes mellitus, symptoms of which include increased blood glucose levels and the appearance of sugar in the urine. The body relies on burning fats, so body weight decreases. However, complete combustion of fat requires the combustion of carbohydrates along with it. Therefore, in diabetes, fat combustion is incomplete and stops at the stage of ketone bodies, such as acetone, which are acidic bodies.

- **Symptoms of Increased Insulin Secretion (Hyperinsulinism):**

The blood glucose level decreases. Glucose is the only fuel for the nervous system, and thus the first system to feel glucose deficiency is the nervous system.

- **Symptoms of Hypoglycemia (Low Blood Sugar):**

- Feeling of intense hunger and fatigue.
- Inability to perform mental effort, followed by dizziness, then coma known as hypoglycemic coma. This is dangerous if not treated within minutes and is treated by injecting glucose intravenously.

The normal blood glucose level ranges between 80 to 120 mg%. It increases after eating, may reach 160-180 mg%, but returns to its original level 2-3 hours after eating. If the blood glucose

level increases to 180 mg%, this is known as the renal threshold for sugar, and sugar appears in the urine.

- **Adrenal Gland (Suprarenal Gland):**

Two small, triangular glands, one located above each kidney.

The adrenal gland consists of two parts that differ in structure and function: the cortex and the medulla.

- **Adrenal Cortex:**

The adrenal cortex is essential for life. It secretes as a result of its stimulation by the hormone from the anterior pituitary lobe (ACTH - Adrenocorticotrophic Hormone). It secretes 3 groups of hormones:

1. Hormones that regulate the levels of sodium and potassium in the blood (Mineralocorticoids, e.g., Aldosterone).
2. Hormones that regulate blood glucose levels (Glucocorticoids, e.g., Cortisol).
3. Hormones similar to sex hormones (Androgens).

- **Adrenal Medulla:**

- The adrenal medulla belongs to the sympathetic nervous system and secretes two hormones:

- **Adrenaline (Epinephrine):** constitutes about 80% of the secretion.
- **Noradrenaline (Norepinephrine):** constitutes about 20% of the secretion.

- **Functions of Adrenaline and Noradrenaline:**

- The adrenal medulla secretes in situations that require increased body resistance, such as:
 - Low arterial blood pressure, as in hemorrhage.
 - Low blood sugar.
 - Exposure to severe cold.
 - During athletic effort.
 - In emotional states.

From this, we see that there is no relationship between the adrenal medulla and the pituitary gland.

The primary sex glands in the female are the ovaries, and in the male, they are the testes. They do not begin their function until puberty.

Testis

The testis is a mixed gland with two types of secretions:

1. **Internal Secretion (Endocrine):**

This is the hormone **testosterone**, which gives the male masculine characteristics such as body hair density, deep voice, and growth of the external genitalia.

2. **External Secretion (Exocrine):**

These are the **spermatozoa**, which represent the reproductive cell in the male and are responsible for fertilizing the ovum and forming the embryo.

The secretions of the testis begin at puberty, around the age of thirteen.

Semen

Semen consists of spermatozoa and secretions from the prostate gland and other glands adjacent to the male reproductive system. These secretions nourish the spermatozoa and help their activity and motility. The volume of semen ranges from 2-5 cc per ejaculation and contains a huge number of spermatozoa, about 100 million per cubic centimeter. A single sperm is sufficient to fertilize the ovum and initiate pregnancy.

Spermatozoa are very active; each has a head, body, and tail that propels it during its movement within the female reproductive tract until one of them meets and fertilizes the ovum.

Spermatozoa are formed in the testes and then stored in the epididymis. During intercourse, the walls of the epididymis and the vas deferens contract, pushing the spermatozoa into the urethra, where they mix with secretions from the prostate gland and the seminal vesicles. Upon ejaculation, semen is expelled from the penis through the external urethral orifice to the outside of the body and into the female reproductive system.

Ovary

The ovary has two important functions:

1. Production of ova (eggs).
2. Secretion of ovarian hormones, which are estrogen and progesterone. The functions of the ovary begin at puberty and are regulated by hormones from the anterior pituitary lobe (L.H. & F.S.H.). The ovary functions for a period of about 30 years and then stops. The cessation of ovarian function is known as menopause.

- **Production of Ova in the Ovary:**

Microscopic Structure of the Ovary:

The ovary is composed of two layers (Cortex and Medulla):

Cortex:

The cortex contains a very large number of ovarian follicles, numbering about 200,000 in the ovary. The follicles represent the ova that the ovary will release during the female's reproductive life.

Among these follicles are cells called "interstitial cells," whose function is to secrete some hormones.

Medulla:

The medulla consists of connective tissue containing blood vessels, lymphatic vessels, and nerves. The ovarian follicle is a spherical body about 45 microns in diameter, containing in its center the ovum, which is about 20 microns in diameter. The ovum is surrounded by two layers of epithelial tissue with fluid between them.

The growth of ovarian follicles begins at puberty and occurs in cycles known as ovarian cycles. Each cycle lasts on average 28 days.

Each ovarian cycle begins with the growth of some ovarian follicles, but only one follicle completes its maturation. The other follicles that began to grow secrete a hormone called

"estrogen," which helps the mature follicle to rupture.

The follicle's diameter reaches 2 mm or more, occupying most of the ovarian cortex and bulging from its surface. The ovum is soon released into the peritoneal cavity.

- **Corpus Luteum:**

A temporary endocrine gland formed in the place of the ovum. The corpus luteum secretes two hormones, "estrogen and progesterone." The corpus luteum forms after the ovum is released from the ovary and begins to grow, reaching its maximum growth around day 22 of the ovarian cycle. If the ovum is not fertilized, the function of the corpus luteum gradually decreases until it atrophies completely.

However, if the ovum is fertilized, the corpus luteum continues to grow until its diameter reaches 2-3 cm and is called the "corpus luteum of pregnancy." Its function continues in the first half of the pregnancy period, then it begins to atrophy, and the placenta replaces it in secreting hormones.

The ovarian cycle lasts 28 days, and ovulation occurs on days 12-14. The process of the ovum's release from the ovary is called "ovulation."

- **Menstrual Cycle:**

The female's arrival at puberty is marked by the periodic discharge of blood accompanied by mucus from the vagina in cycles known as menses or the menstrual cycle.

The menstrual cycle lasts 28 days, but it can be shorter (25 days) or longer (31 days), provided it is regular. The menstrual cycle begins at puberty and stops at menopause. It also stops during pregnancy and sometimes during lactation.

The menstrual cycle is divided into 3 phases:

1. **Menstrual Phase (Breakdown - Blood Flow):**

In this phase, the artery supplying the uterine mucosa constricts, reducing the amount of blood flowing through it, and the surface cells die. This dead tissue, along with an amount of blood ranging from 50 to 150 cc, passes to the outside of the body.

2. **Proliferative Phase (Growth and Multiplication):**

This phase extends from the end of the menstrual phase until day 14 of the cycle, i.e., when the ovum is released from the ovary.

In this phase, the cells of the uterine mucosa begin to grow, the length of the uterine glands increases, and they become more coiled. Also in this phase, the growth of the uterine muscles increases, and the amount of blood passing through it increases. This phase occurs under the influence of estrogen hormone.

3. **Secretory Phase:**

Lasts from day 15 of the cycle until its end. The growth of the glands increases, and they become filled with secretion. This phase occurs under the influence of both estrogen and progesterone hormones.

- **Ovarian Hormones:**

1. **Estrogen:**

Estrogen is the hormone responsible for the physical and psychological changes that occur in the female at puberty and maintains them until menopause.

2. **Progesterone:**

Helps regulate the menstrual cycle and increases uterine activity, preparing it for pregnancy.

- 3.

QUESTIONS:

1. Mention the hormones of the anterior pituitary lobe.

2. What do you know about:

- Tetany disease.
- Simple goiter.
- Myxedema.
- Symptoms of low blood sugar (hypoglycemia).

3. What happens if:

A- Growth hormone secretion increases before puberty?

B- Thyroxine hormone secretion decreases before puberty?

C- Parathormone secretion increases?

4. Complete the following:

A- The posterior lobe of the pituitary gland secretes two hormones:

B- The thyroid gland secretes a hormone called and its function is

C- Parathormone is a hormone secreted by the gland and is responsible for regulating in the blood.

D- The adrenal cortex secretes hormones to regulate

E- Progesterone is a hormone secreted by and helps with

5. Choose the correct answer from the following:

A) Tetany disease arises from a deficiency of:

() Calcium ions in the blood.

() Thyroxine hormone.

() Growth hormone.

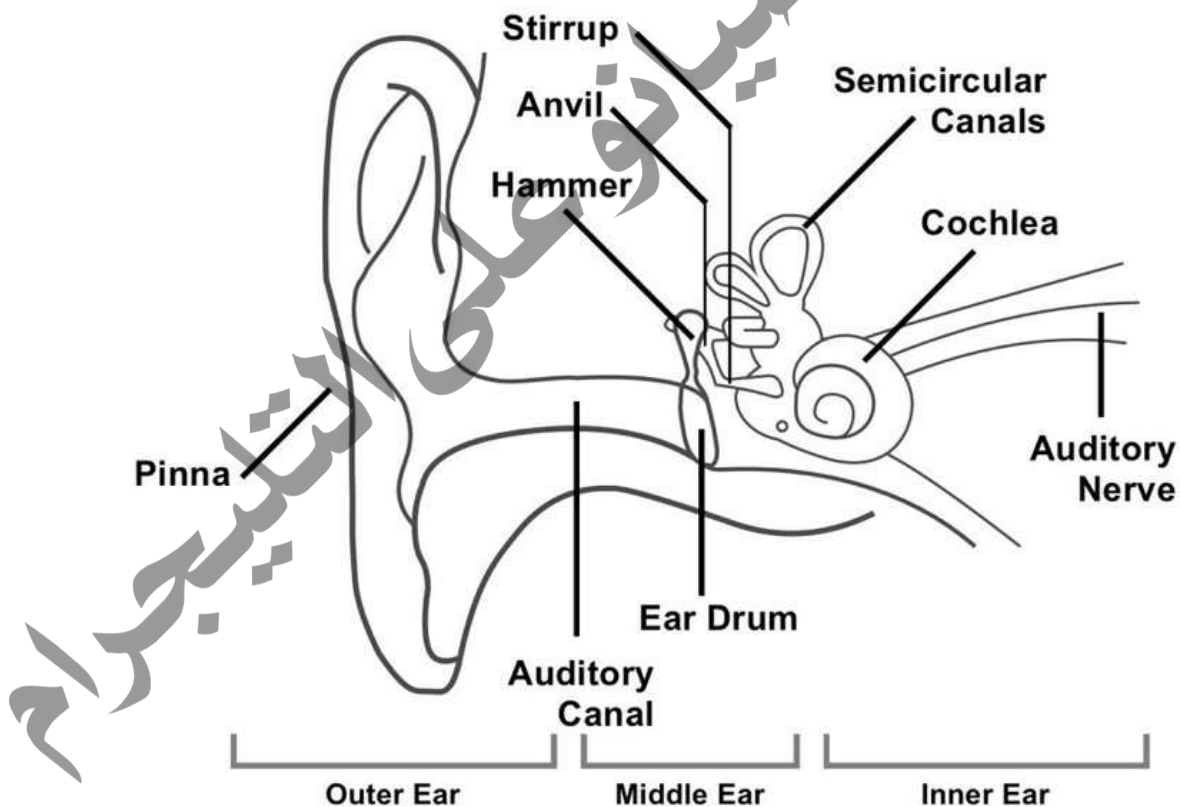
() Insulin hormone.

SENSE ORGANS

Sense of Hearing

Hearing is the sense of perceiving sound through the ear, which is adapted to perform this function in addition to maintaining balance.

- The ear is composed of three main parts:
 - External Ear • Middle Ear • Internal Ear
 - **External Ear:** Consists of:
 - **Pinna (Auricle):** Works to collect sounds and direct them into the auditory canal.
 - **Auditory Canal (External Auditory Meatus):** About 2.5 cm long. It secretes a waxy substance to which dust and germs adhere, preventing their entry into the ear.
 - **Tympanic Membrane (Eardrum):** Separates the external ear from the middle ear.
 - **Middle Ear:** Contains three small bones (ossicles):
 - **Malleus (Hammer) - Incus (Anvil) - Stapes (Stirrup)**
- The middle ear connects to the pharynx via the Eustachian tube, which works to equalize pressure on both sides of the eardrum.



- **Internal Ear:** Consists of:

- **Cochlea:** Known as the hearing apparatus.
- **Semicircular Canals:** Known as the balance apparatus.

- **Mechanism of Hearing:**

- (1) Sound travels in the air in all directions in the form of waves, originating from the sound source, reaching the ear in a fraction of a second.
- (2) The pinna collects the sound waves and directs them to the eardrum via the auditory canal.
- (3) The eardrum vibrates when sound waves hit it – causing the three ear ossicles to vibrate back and forth – which transmit and amplify the sound waves.
- (4) The oval window membrane vibrates according to the vibration of the three ossicles, moving the fluid in the cochlea.
- (5) The movement of the cochlear fluid stimulates the sensory cells to send nerve impulses via the auditory nerve to reach the auditory center in the cerebral hemispheres of the brain, where the person perceives and distinguishes the sound.

- **Sense of Sight (Vision):**

- The sense of sight is the sense of perceiving images and colors through the eye, which is adapted to perform this function.
- The eye rests in a cavity in the skull called the eye socket (orbit).

- **Structure of the Eye:**

The eyeball has a wall of three layers (Sclera – Choroid – Retina):

- **Sclera:**

This is the white of the eye. Its front part is transparent and is called the cornea.

- **Iris:**

This is the colored part of the eye. In its center is a hole called the pupil.

- **Lens of the eye:**

Biconvex, located behind the iris. In front of it is the aqueous humor, and behind it is the vitreous humor.

- **Retina:**

Contains millions of sensory cells; therefore, it is responsible for vision. The most sensitive area is the macula lutea (center of vision). In contrast, the blind spot is located at the point where the optic nerve exits the eye and does not perceive vision. The retina contains rod cells, which do not distinguish colors, and cone cells, which do distinguish colors.

- **Choroid:**

A black tissue located behind the retina and before the sclera. It absorbs light and prevents its reflection outside the eye. It supplies the retina with nourishment and oxygen.

- **Protection of the Eye:** Provided by:

- Skull bones (orbit)
- Eyelashes
- Lacrimal glands (tear glands)
- Conjunctival membrane
- Eyelids

- **Mechanism of Vision:**

The eye's function resembles that of a camera.

1. Light is reflected off visible objects to fall on the eye.
2. Light rays are refracted as they pass through the cornea and then the lens.
3. The refracted rays converge on the retina, which contains light-sensitive cells, forming a small, inverted image of the visible object.
4. When the image is carried to the brain via the optic nerve, it is corrected to become upright and match the object in its natural position.

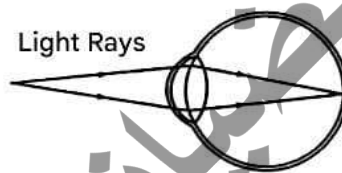
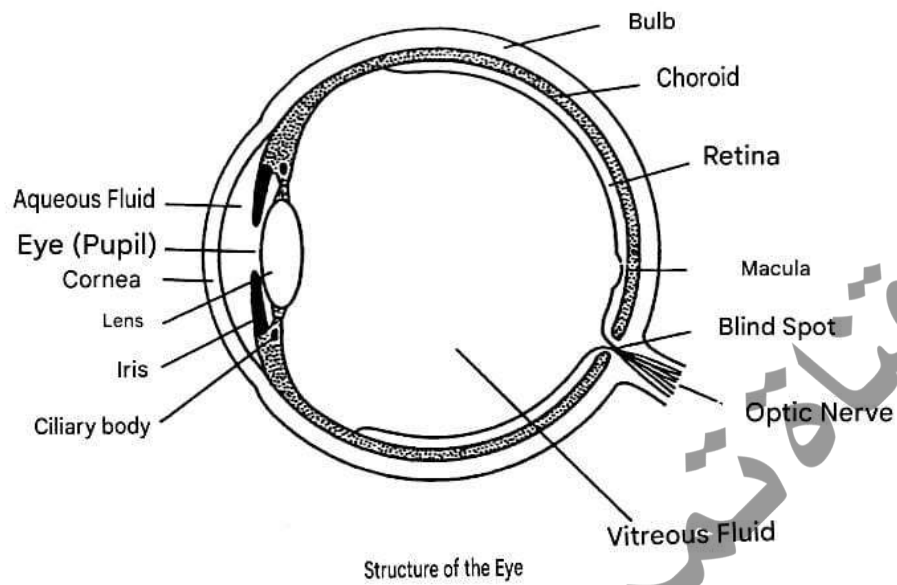
- **Accommodation of the Eye:**

1. **Regarding the amount of light entering the eye:**

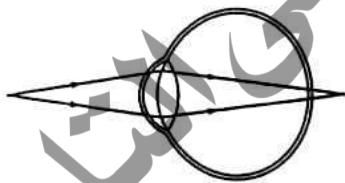
- In bright light: The pupil constricts to reduce the amount of light entering the eye.
- In dim light: The pupil dilates to allow more light to enter the eye.

2. **Regarding the distance of objects from the eye:**

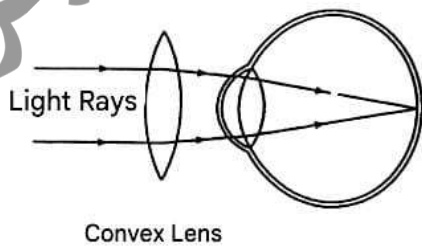
Some objects may be near or far so that their image does not fall exactly on the retina, making vision unclear. Therefore, the curvature of the lens is changed by the ciliary muscles.



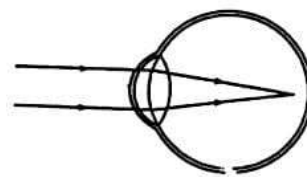
(2) Hyperopia



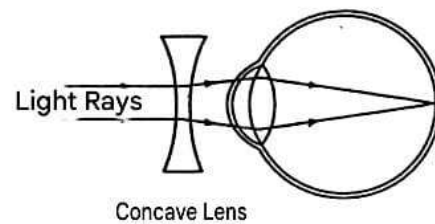
Hyperopia is treated
by using glasses with convex lenses



(1) Myopia



Myopia is treated
by using glasses with concave lenses



- **Vision Defects:**

A healthy person sees objects clearly within a range of 25 cm to 6 meters.

- **A/ Myopia (Nearsightedness):**

In this condition, a person sees nearby objects (at a distance of 25 cm) clearly, while distant objects have their image formed in front of the retina and are unclear. Myopia is treated using concave lenses.

- **B/ Hyperopia (Farsightedness):**

In this condition, a person sees nearby objects unclearly, as their image is formed behind the retina. Hyperopia is treated using convex lenses.

- **Sense of Taste:**

- The tongue is the organ of taste.
- Various taste buds of different shapes and sizes are scattered on the surface of the tongue. Each taste bud contains sensory nerve cells that can distinguish one or two of the following tastes:
 - Sweet
 - Salty
 - Sour
 - Bitter
- A prerequisite is that these substances must be dissolved in the layer of saliva that covers the tongue.
- Nerve signals are transmitted from the taste buds via the seventh cranial nerve (from the front two-thirds of the tongue) and the ninth cranial nerve (from the back third of the tongue) to the taste centers in the brain, where the taste of food is perceived.

- **Sense of Smell:**

- The nose is the organ of smell and odor perception. It is a cartilaginous organ that begins with the nostril opening leading to the nasal cavity.
- The roof of the nasal cavity is lined with a layer of cells containing (chemical) sensory receptors connected to fibers of the olfactory nerve, which ends at the smell centers in the brain.
- A prerequisite for a substance to be smelled is that it must be volatile or soluble in the mucous layer that covers the roof of the nasal cavity.
- When these substances contact the membrane lining the olfactory region, the chemical receptors are stimulated to release nerve signals via the fibers of the olfactory nerve to the smell centers in the brain, where the odor is perceived.

- **Sense of Touch:**

- The skin is the organ responsible for the sense of touch.
- The dermis layer contains many nerve endings that gather in the form of papillae, responding to different stimuli such as:
 - Touch – Pressure – Heat – Cold – Pain.

QUESTIONS:

1. What are the main parts of the ear? Explain the function of each.
2. Explain the steps of sound perception (hearing).
3. Draw and label the structure of the eye.
4. Explain the mechanism of vision.

5. Explain:
A / How the amount of light entering the eye is adjusted.
B / The role of the eye's lens in seeing objects.

6. Explain vision defects.
7. Talk about:
A / The sense of taste.
B / The sense of smell.

8. Complete the following:
A / The Eustachian tube works to
B / The eye is protected by

9. Choose the correct answer from the following:
A / The amount of light entering the eye is adjusted by the:
() Cornea.
() Pupil.
() Lens.
() Retina.